



GS FOUNDATION 2024-25

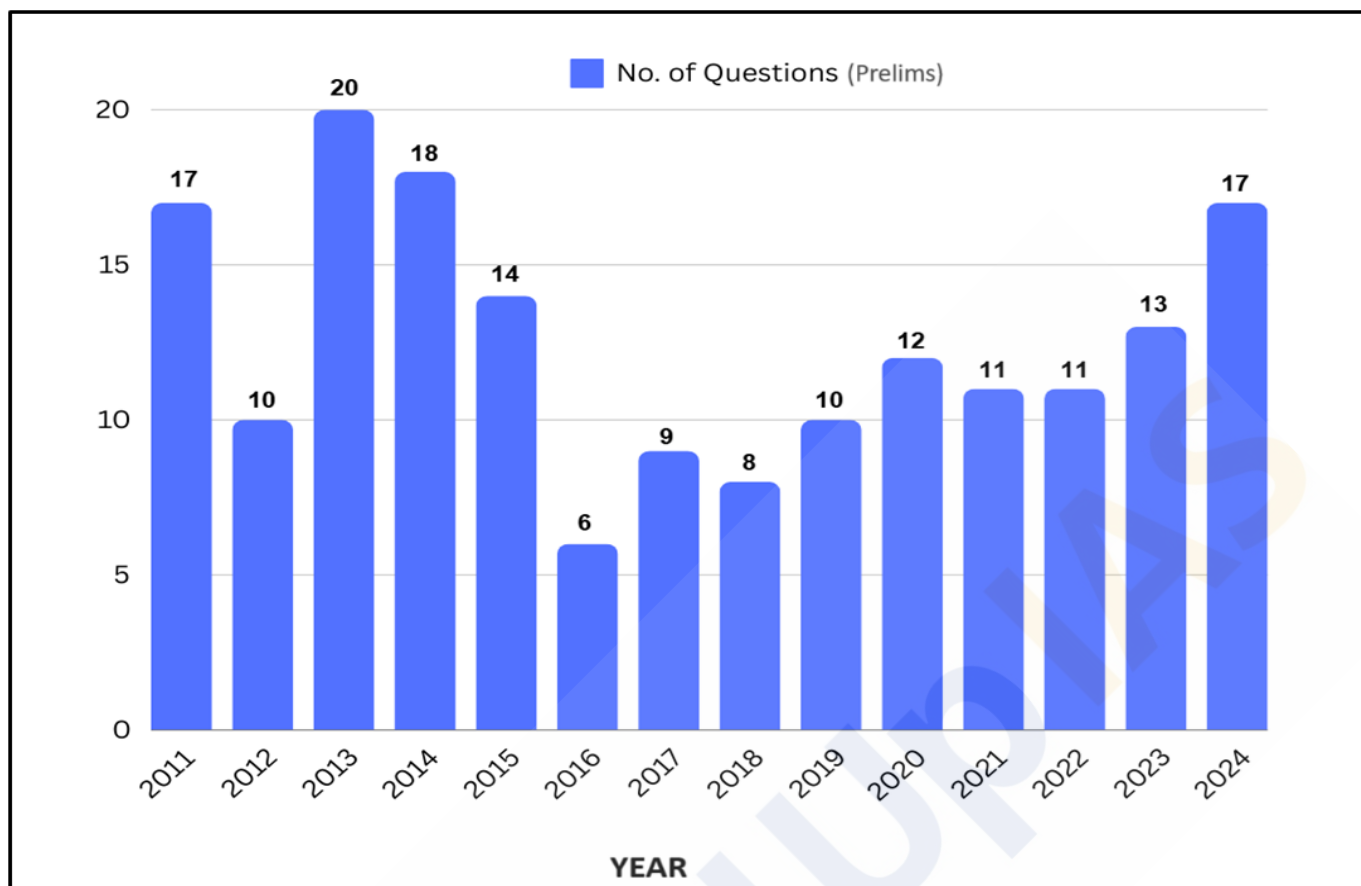
GEOGRAPHY - 22

Soil

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Soil

Introduction:

Soil is the **uppermost layer of the Earth's crust**, formed by the continuous weathering of mountains over thousands of years. It is composed of **four basic constituents - minerals, organic materials, air, and water**. The texture of soil is determined by the proportions of **sand, silt, and clay it contains**. Organic matter, such as leaves, decomposes to form an upper organic layer known as humus, which is crucial for soil fertility.

Soil Genesis:

- Soils play a vital role in the natural environment, linking climate and vegetation, and significantly impacting human activities through their fertility. The scientific study of soils is called **pedology**, and the process of soil formation is known as **pedogenesis**. Soil is the upper weathered layer of the Earth's crust, influenced by plants and animals. A vertical section through this zone constitutes a **soil profile**, which typically consists of several distinguishable layers or horizons, each representing different types of soil.
- Soil is a relatively thin surface layer of mineral matter that normally contains a significant amount of organic material and is capable of supporting living plants. It occupies the outer skin of the Earth, extending from the surface down to the maximum depth reached by living organisms, primarily plant roots.
- **Soil's ability to produce and store plant nutrients is made possible by the interactions of diverse factors such as water, air, sunlight, rocks, plants, and animals.** Although thinly distributed over the land surface, soil functions as a fundamental interface where the atmosphere, lithosphere, hydrosphere, and biosphere meet.

Soil Development:

- The development of soil begins with **the physical and chemical disintegration** of rock exposed to the atmosphere and the action of water percolating from the surface, a process called **weathering**. Weathering weakens and breaks down solid rock, creating loose inorganic material called regolith, which blankets the unfragmented rock below. The upper portion of the regolith, typically about half a meter deep, differs from the material below due to **more intense biological and chemical processes**.
- This upper portion is soil, composed largely of finely fragmented mineral particles, living plant roots, decaying plant parts, microscopic organisms, and variable amounts of air and water. **Soil is not the end product of a process but a stage in a continuous cycle of physical, chemical, and biological processes.**

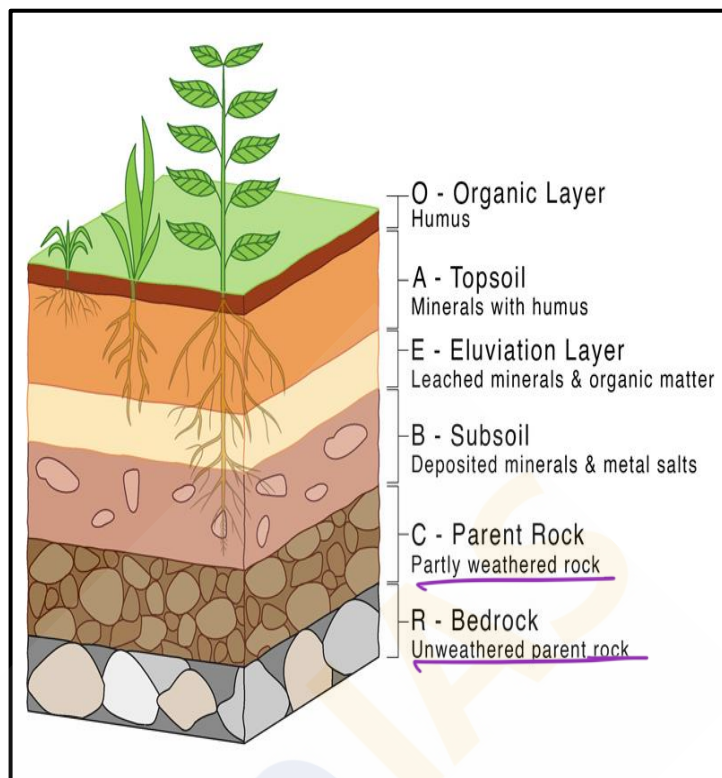
Soil Profile:

There are different types of soil, each with its own set of characteristics. Dig down deep into any soil, and you'll see that it is made of layers, or **horizons (O, A, E, B, C, R)**. Put the horizons together, and they form a soil profile. Like a biography, each profile tells a story about the life of a soil. Most soils have three major horizons (A, B, C) and some have an organic horizon (O).

- **(Humus or organic)** - Mostly **organic matter** such as decomposing leaves. The O horizon is thin in some soils, thick in others, and not present at all in others.
- **A (Topsoil)** - Mostly minerals from **parent material with organic matter incorporated**. A good material for plants and other organisms to live.

moving down

- **E (Eluviated)** - Leached of **clay, minerals, and organic matter**, leaving a concentration of sand and silt particles of quartz or other resistant materials – missing in some soils but often found in older soils and forest soils.
- **B (Subsoil)** - Rich in minerals that leached (moved down) from the A or E horizons and accumulated here. *partially deposit rock*
- **C (parent material)** - The deposit at Earth's surface from which the soil developed.
- **R (Bedrock)** - A mass of rock such as **granite, basalt, quartzite, limestone** or **sandstone** that forms the parent material for some soils – if the bedrock is close enough to the surface to weather. This is not soil and is located under the C horizon.



Soil-Forming Processes:

Soil formation involves four main processes - **soil enrichment, removal, translocation, and transformation**. Each process contributes to the development and characteristics of soil in different ways.

Soil Enrichment:

- Soil enrichment involves the **addition of organic or inorganic matter to the soil**. For example, river floods can deposit silt on the soil surface, or wind can blow dust onto it. Organic enrichment occurs as water carries humus from the uppermost layer (O horizon) into the layer below it (A horizon). *O → A*

Removal Processes:

- Removal processes involve the **loss of material from the soil body**. This can occur through erosion, which carries soil particles into streams and rivers, or leaching, where soil compounds and minerals dissolve in water and move to lower levels. **Cheluviation, the downward movement of materials due to organic agents**, is similar to leaching but involves plant acids rather than water alone.

Translocation Processes:

Translocation involves the movement of materials within the soil, either upward or downward.

- **Downward Translocation – 'Eluviation'** occurs **when fine particles like clays and colloids move downward, forming the E horizon**, while sand or coarse silt remains. These materials then accumulate in the B horizon through illuviation, evident in soil profiles: the humus-rich A horizon is brownish, the E horizon is white from colloid removal, and the B horizon is orange-red from **iron sesquioxides**.
 - In **moist climates**, **excess water leaches calcium carbonate from the soil, creating acidic soils**. Adding lime can correct this. In **dry climates**, **insufficient precipitation deposits calcium carbonate in the B horizon, forming a cemented hardpan that impairs soil fertility**. In colder climates, a pan of iron and aluminum oxides can block drainage and keep the soil saturated.
- **Upward Translocation** - In **desert climates**, upward translocation can occur when **groundwater, rich in dissolved salts, is drawn up to the soil surface through capillary tension as surface water evaporates**. This process, known as **'salinization'**, results in the accumulation of salts, which are toxic to many plants. In irrigated desert lands, salinization can ruin the soil with little hope of revival.

Transformation Processes:

- Transformation involves the **alteration of materials within the soil**. Examples include the conversion of primary minerals to secondary types and the decomposition of organic matter by microorganisms to produce humus, a process known as humification.
 - Example - In warm, moist climates, organic matter can be nearly completely transformed into carbon dioxide and water, leaving little organic material in the soil.

Soil Forming Factors: *climate, Parent Material, Time, Topology, Biota*

Climate: → *Dominant Processes* → Dry/Desert → Physical weathering → Coarse Soil
 Humid Reg- → chemical weath. → fine soil

- Climate, encompassing **precipitation and temperature**, plays a crucial role in determining soil properties. **Precipitation controls the downward movement of nutrients and other chemical compounds** through a process called **translocation**. High precipitation levels cause water to wash nutrients deeper into the soil, making them inaccessible to plant roots. Conversely, low precipitation leads to the buildup of salts in the soil, restricting fertility.
- Temperature significantly affects the chemical development of soils and the formation of soil horizons**. Biological activities slow down below 10°C and cease at or below the freezing point (0°C or 32°F), rendering chemical processes inactive. Consequently, **decomposition is slow in cold climates**, resulting in a thick accumulation of organic matter in the O horizon, which then enriches the A horizon. In contrast, in **warm, moist climates of low latitudes**, **bacteria rapidly decompose plant material, leading to a lack of O horizons and minimal organic matter throughout the soil profile**.
 - Black Earth* → cold region → ss. chernozem soil Russia. → most fertile soil
 - Hot region → equatorial soil
- Role of Precipitation** - In regions with high rainfall, **water percolating through the soil leaches nutrients and organic matter from the upper layers, unless counteracted by plant roots**.
 - For instance, **tropical rainforest soils are nutrient-poor due to intensive leaching from heavy rains, with most nutrients stored in the vegetation**. Conversely, arid regions with low annual precipitation experience high evaporation rates, leading to salt accumulation in the soil.
- Role of Temperature** - Solar energy, reflected in **temperature**, influences the form of water on the soil surface and within the soil, **accelerating reactions such as chemical processes, evapotranspiration, and biological activities**. Wide temperature fluctuations, especially with the presence of water, cause soil to shrink and swell, leading to frost action and general weathering.
 - For example, **laterite soils are found in regions with alternating wet and dry climates**. In Rajasthan, high temperatures and wind erosion result in sandy soil formation from both granite and sandstone.

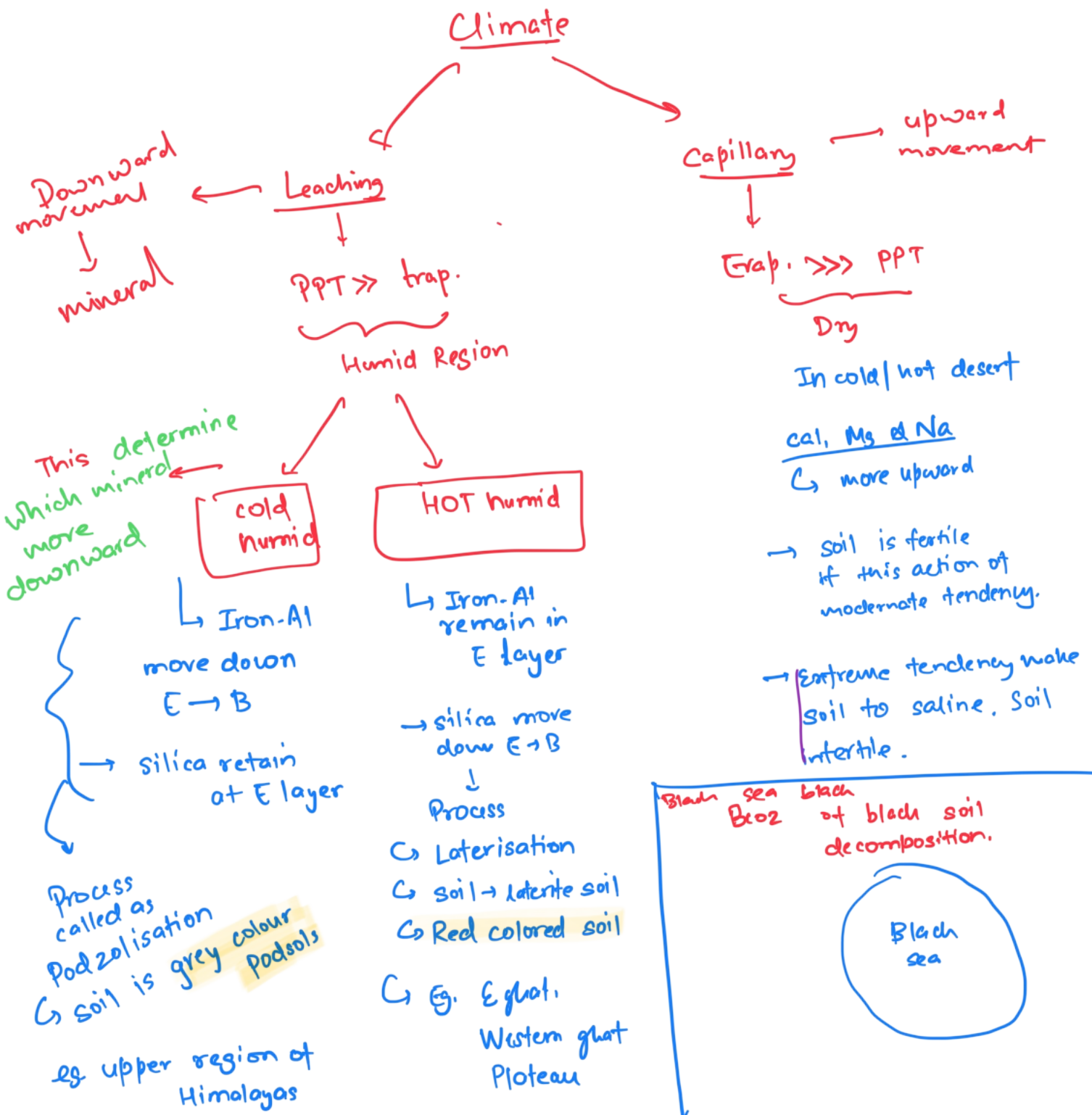
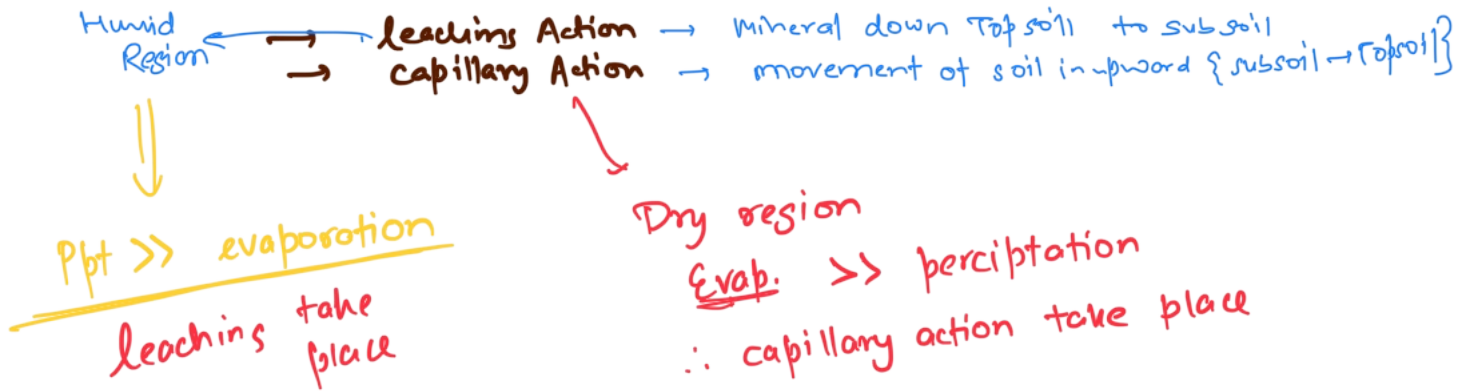
Organisms:

- Living plants and animals**, along with their nonliving organic products, significantly impact soil. **Plant roots mix and disturb the soil while providing organic material to upper soil horizons**. Soil-dwelling organisms, ranging from bacteria to burrowing mammals, further influence soil structure. Earthworms, for example, enhance soil structure, fertility, and profile depth through their burrowing and digestion activities. The presence of well-nourished earthworms often indicates productive soil.

Relief: → *Topology*

- The **shape and configuration of the ground surface**, known as relief, influence soil formation. **Generally, soil horizons are thicker on gentle slopes and thinner on steep slopes due to rapid erosion on steeper inclines**. **Slopes facing away from the Sun tend to have cooler, moister soils, while those facing toward the Sun experience higher temperatures and increased evapotranspiration**. **Topography redistributes water, creating wetter conditions in lowlands and affecting soil processes, distribution, and vegetation types**.
 - China side Himalayas*
 - Sun Ka Kam*

Climate other roles



or permeable (ability to allow water to pass through it) → Red soil/sandy

Parent Material:

- The original source of the parent material influences soil chemistry. For instance, **iron-rich bedrock produces iron oxide-rich soils, while limestone forms calcium-rich soils.** Secondary minerals weathered from primary minerals can impart unique properties to soils. **Soil texture is also determined by the size of mineral grains in the parent material.** Soils inherit properties such as mineral composition, color, particle size, and chemical elements from their parent material. **Basalt → black colored, granite → red colored soil, Limestone → white or light color soil.**
 - For example, **peninsular soils reflect their parent rock, with ancient crystalline and metamorphic rocks like granite, gneiss, and schist forming red soils due to iron oxide content.** Lava rocks produce black-colored soils, and sandstone yields sandy soils. Soils in the northern plains, transported and deposited from Himalayan and peninsular sources, have little relation to the in-situ rock material.

This tells about water holding property

Time:

Texture

Basalt → fine texture
Granite → coarse texture

② Mineral composition

Limestone → Calcium rich
Basalt → Ferro Mg. rich

- The development of soil characteristics and properties requires time. **A fresh deposit of mineral matter, like clean, sorted sand from a dune, may take hundreds to thousands of years to develop into structured soil.** Soil-forming processes are typically slow, with thin soil layers taking centuries to form on newly exposed surfaces. Warm, moist environments accelerate soil development, while parent material attributes play a significant role. For example, soil develops relatively quickly from sediments but slowly from bedrock. **A common rule of thumb is that it takes about 500 years to form 2.5 cm (1 in.) of topsoil.** → **Mature soil has well developed horizon.**

Human Activity:

- Human activities also influence soil's physical and chemical nature. **Clearing native vegetation for crops can induce erosion, removing upper layers rich in organic matter.** Long-term agricultural practices have significantly altered soil structure and composition, creating distinct soil classes that are just as important as natural soils.

Classification of Soils in India:

India's diverse relief features, landforms, climatic conditions, and vegetation types have played a significant role in the formation and development of various soil types across the country. These soils vary widely in their characteristics and have been historically classified based on different criteria.

In ancient times, soils in India were broadly categorized into **two main groups** – 'Urvara' and 'Usara.' The **Urvara soils were considered fertile, while the Usara soils were deemed sterile.** This simple classification was used to distinguish between productive and non-productive lands.

By the 16th century, soil classification in India became more detailed. Soils were categorized based on their **inherent characteristics and external features such as texture, colour, land slope, and moisture content.** Texturally, soils were identified as sandy, clayey, silty, and loamy. In terms of colour, soils were described as red, yellow, or black, among other variations.



Since India gained independence, scientific surveys and studies of soils have been conducted by various organizations. The Soil Survey of India, established in 1956, undertook extensive soil studies, particularly in areas like the Damodar Valley. The National Bureau of Soil Survey and Land Use Planning, an institute under the Indian Council of Agricultural Research (ICAR), also conducted comprehensive studies on Indian soils.

In an effort to align soil studies with international standards, ICAR classified Indian soils according to the United States Department of Agriculture (USDA) Soil Taxonomy. This classification system takes into account the nature and characteristics of soils, making it comparable on a global scale.

ICAR has classified the soils of India into the following order as per the USDA soil taxonomy

Sl. No.	Order	Area (in Thousand Hectares)	Percentage
(i)	Inceptisols	130372.90	39.74
(ii)	Entisols	92131.71	28.08
(iii)	Alfisols	44448.68	13.55
(iv)	Vertisols	27960.00	8.52
(v)	Aridisols	14069.00	4.28
(vi)	Ultisols	8250.00	2.51
(vi)	Mollisols	1320.00	0.40
(viii)	Others	9503.10	2.92
Total			100

Source : Soils of India, National Bureau of Soil Survey and Land Use Planning, Publication Number 94

Classification Based on Soil Characteristics:

The soils of India have been classified into the following categories based on their **genesis, colour, composition, and location** - Alluvial Soils, Black Soils, Red and Yellow Soils, Laterite Soils, Arid Soils (Desert Soils), Forest Soils (Mountain Soils), Saline Soils and Peaty Soils.

Alluvial Soils:

Alluvial soils are primarily formed due to the **deposition of silt by the Indo-Gangetic-Brahmaputra rivers**. Additionally, in **coastal regions, wave action** contributes to the formation of some alluvial deposits. The **parent material** of these soils is derived from the **rocks of the Himalayas**, making it of transported origin. Alluvial soils represent the largest soil group in India, covering approximately 15 lakh square kilometers, which accounts for about **46% of the country's total area**. These soils are highly productive, **supporting more than 40% of India's population** by providing fertile agricultural land.

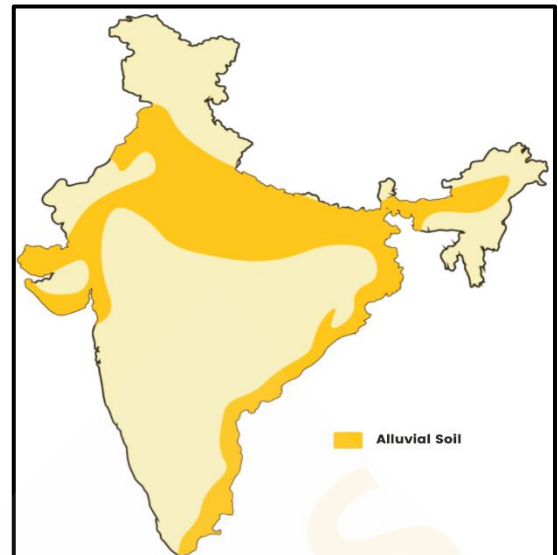
Characteristics:

- Alluvial soils are relatively **young and immature, with weakly developed soil profiles** due to their recent origin. The soil texture varies widely, ranging from **sandy to clayey, with sandy soils being predominant**. In drier regions, the soils tend to be loamy to sandy-loam, while in deltaic areas, they exhibit a clayey loam texture. Pebbly and gravelly soils are rare in these regions, although beds of **Kankar** (calcareous concretions) are present along some river terraces.
- The **loamy nature** of alluvial soils imparts **good composition** sand, silt & clay **porosity**, allowing good drainage and creating conditions **favourable for agriculture**. Due to the **recurrent flooding** in these areas, the **soils are constantly replenished** with new layers of alluvium, maintaining their fertility.
- The chemical composition of alluvial soils is characterized by a generally **low proportion of phosphorous and nitrogen**. However, they contain **adequate amounts of potash and alkalies**. The levels of iron oxide and lime in these soils vary widely depending on the specific location.



Distribution:

- Alluvial soils are predominantly found along the **Indo-Gangetic-Brahmaputra plains**, except in regions where the top layer is covered by desert sand. Additionally, these soils are present in the **deltas of the Mahanadi, Godavari, Krishna, and Cauvery rivers**, where they are referred to as deltaic or coastal alluvium. Smaller patches of alluvial soils are also found in the **Narmada and Tapi valleys** and in the northern parts of Gujarat.

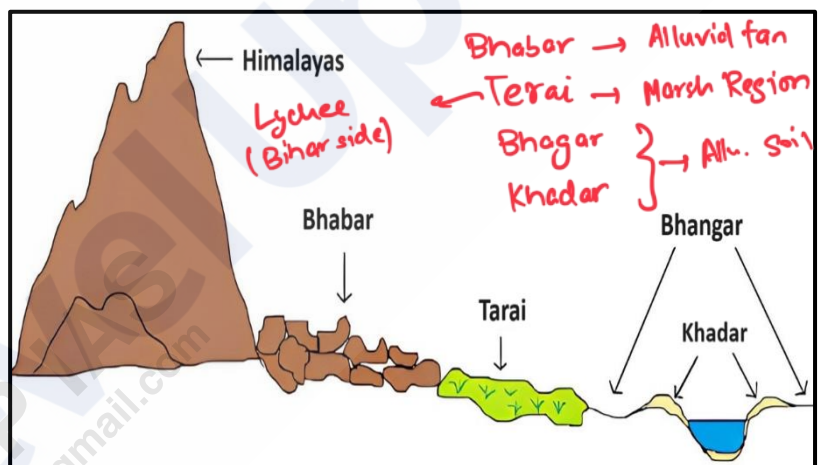


Agricultural Importance:

- Due to their flat and regular surface, alluvial soils are **highly suitable for agriculture**. These soils respond well to irrigation, including canal and well/tube-well irrigation, and are capable of yielding a wide variety of crops. Crops that thrive in alluvial soils include **rice, wheat, sugarcane, tobacco, cotton, jute, maize, oilseeds, vegetables, and fruits**.

Geological Divisions:

- Geologically, the alluvial soils of the Great Plain of India are divided into **two main types** - the newer or younger **Khadar** and the older **Bhangar** soils. Additionally, the **Bhabar** and **Terai** regions represent distinct belts within the alluvial plains.
- Bhabar Region** - The Bhabar belt is a **porous stretch**, about 8-16 kilometers wide, located along the **foothills of the Shiwaliks** in the **northernmost part of the Indo-Gangetic plain**. Here, rivers descending from the Himalayas deposit their load in the form of alluvial fans, which have merged together to form the Bhabar belt. The unique feature of this region is its porosity, resulting from the **accumulation of pebbles and rock debris** across the alluvial fans. Due to this porosity, **streams tend to disappear once they reach the Bhabar region, leaving dry riverbeds except during the rainy season**. The region is **not suitable for agriculture**, and only large trees with deep roots can thrive here.
- Terai Region** - South of the Bhabar belt lies the Terai, a **poorly drained, marshy, and thickly forested narrow tract**, approximately 15-30 kilometers wide. The **underground streams from the Bhabar belt re-emerge in the Terai**, creating a swampy lowland with silty soils. **Terai soils are rich in nitrogen and organic matter but deficient in phosphate**. While the region is typically covered by tall grasses and forests, it is **suitable for cultivating crops** such as **wheat, rice, sugarcane, and jute**. The dense forests in this area provide habitat for diverse wildlife.
- Bhangar Soils** - Bhangar represents the **older alluvium** that forms terraces along riverbeds, typically lying about 30 meters above the **floodplain**. These soils have a more **clayey composition** and are generally **darker in colour**. Beneath the terrace surface of the Bhangar soils, beds of lime nodules known as "**Kankar**" are often found.
- Khadar Soils** - Khadar soils are composed of **newer alluvium** and form the **floodplains along riverbanks**. These areas are **frequently flooded, leading to the deposition of a new layer of alluvium** with each flood, making Khadar soils the **most fertile** in the Ganges basin. Khadar soils are typically **sandy clays and loams**, characterized by their drier and leached nature. They are less calcareous and carbonaceous, with **fewer kankar nodules, compared to Bhangar soils**.



Black Soils (Regur):

Black soils, also known as **regur soils** or **black cotton soils**, are primarily formed from the **weathering of basaltic rocks that emerged during the fissure eruptions of the Cretaceous period**. These **volcanic rocks**, predominantly found in the **Deccan Plateau** (including the Deccan and Rajmahal traps), serve as the **parent material** for most black soils. However, in Tamil Nadu, gneisses and schists act as the parent material, with gneisses typically forming deep soils and schists forming shallower soils. The regions where black soils are found are characterized by high temperatures and low rainfall, making this soil type typical of the dry and hot regions of the Indian Peninsula.

Black soils **cover more than 15% of India's land area**. Their **distinctive black colour** is attributed to the presence of titaniferous magnetic compounds found in basalt.

Characteristics:

- A typical black soil is highly argillaceous, containing a **significant proportion of clay, often 62% or more**. These soils exhibit considerable variation in fertility depending on their location. While the **black soils found in upland areas are generally of low fertility, those located in valleys are highly fertile**.
- One of the defining characteristics of black soils is their **high moisture retention capacity**. **When wet, these soils swell significantly, becoming very sticky and challenging to work with during the rainy season**. Conversely, in the summer, as **moisture evaporates, the soil shrinks and develops broad, deep cracks**. Despite this, the **lower layers of the soil can retain moisture**, which contributes to the **soil's extraordinary fertility**. The cracks formed during the dry season also **allow for oxygenation to penetrate to sufficient depths, further enhancing soil fertility**. This property, along with the soil's tendency to develop a blocky structure when dry, is referred to as its **"self-ploughing" capacity**.
- The **black colour** of these soils is primarily due to the **presence of titaniferous magnetite or iron**, along with other black constituents of the parent rock. In certain regions, such as Tamil Nadu and parts of Andhra Pradesh, the black colour is derived from **crystalline schists and basic gneisses**. The black soils exhibit various shades of black, ranging from deep black to medium black, shallow black, and even mixtures of red and black.
- Chemically, the black soils are **rich in lime, iron, magnesia and alumina**. They also **contain potash**. But they **lack in phosphorous, nitrogen and organic matter**. The colour of the soil ranges from deep black to grey.

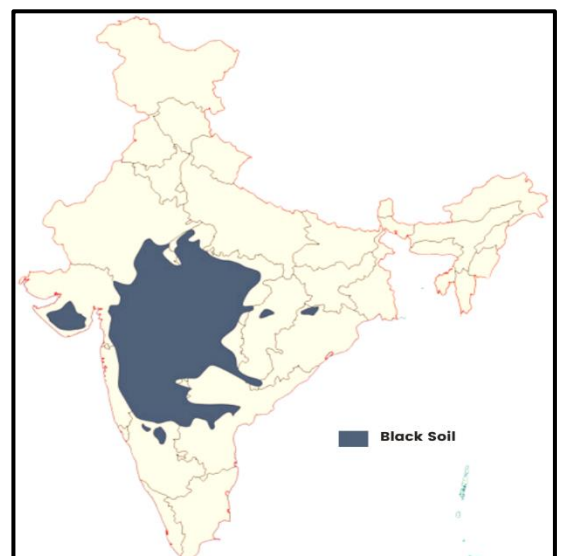


Distribution:

- Black soils are predominantly found in the **Deccan lava plateau region of India**, covering an area of approximately 46 lakh square kilometers, which **constitutes 16.6% of the country's total area**. These soils are spread across the states of **Maharashtra, Madhya Pradesh**, parts of **Karnataka, Telangana, Andhra Pradesh, Gujarat**, and **Tamil Nadu**.

Agricultural Importance:

- Black soils are particularly **well-suited for cotton cultivation**, which is why they are often referred to as black cotton soils. In addition to cotton, these soils support the growth of a variety of other crops, including **wheat, jowar (sorghum), linseed, Virginia tobacco**,



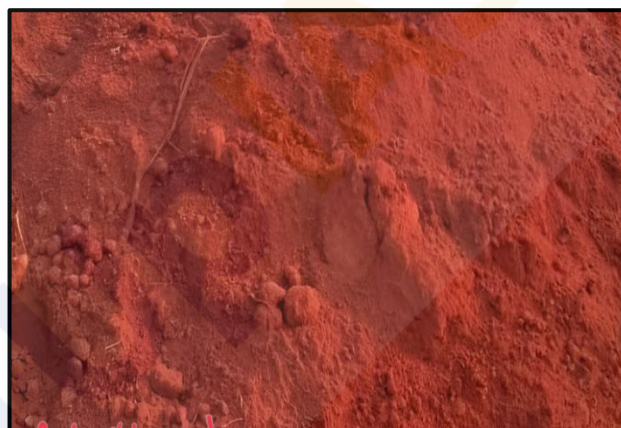
castor, sunflower, and millets. Where irrigation facilities are available, rice and sugarcane are also significant crops. Furthermore, black soils are capable of supporting the successful cultivation of a wide range of vegetables and fruits. Remarkably, these soils have been used for growing crops for centuries without the need for added fertilizers or manures, with little to no evidence of soil exhaustion.

Red Soils: → Permeable soil → Dry crops → Millets, pulses, oil seeds

Red soil, which develops primarily on **Archean granite**, is a significant soil type in India, covering the **second-largest area of the country**. The distinct red coloration of this soil is due to the presence of **ferric oxides**, which form thin coatings on the soil particles. The **top layer** of the soil appears red, while the horizon beneath it is typically yellowish. This soil type covers approximately **18.5% of India's total land area** and exhibits a texture that ranges from sandy to clayey and loamy. It is often referred to as the "**omnibus group**" due to its **widespread occurrence** and varied characteristics.

Characteristics:

- Red soils exhibit a wide range of characteristics influenced by the variability in rainfall across different regions. This variability has led to the development of three subtypes:
 - Red and Yellow Soil** - Found in regions receiving **heavy rainfall of up to 200 cm**, such as the northeastern parts of India, including **Nagaland, Mizoram**, and the hills of **Manipur**, as well as parts of the **Malabar coast**. Due to the high rainfall, these soils require quick drainage to avoid waterlogging. *Not majorly found as rice area have laterite soil.*
 - Red Sandy Soil** - Predominantly found in **drier plateau** regions like **Karnataka, Tamil Nadu, Telangana**, and **Rajalaseema**, where annual **rainfall ranges from 40 to 60 cm**. These soils are typically well-drained and sandy in structure.
 - Red Alluvial Soil** - Located along river valleys, this subtype has **relatively higher fertility** compared to the other subtypes. These soils support a variety of crops due to their enhanced fertility and favourable physical properties.
- Red soils are **well-drained and have a sandy structure**, making them particularly **rich in iron and potash**. However, they are **deficient in** several essential nutrients, including **phosphate, lime, magnesia, humus, and nitrogen**, which limits their natural fertility.

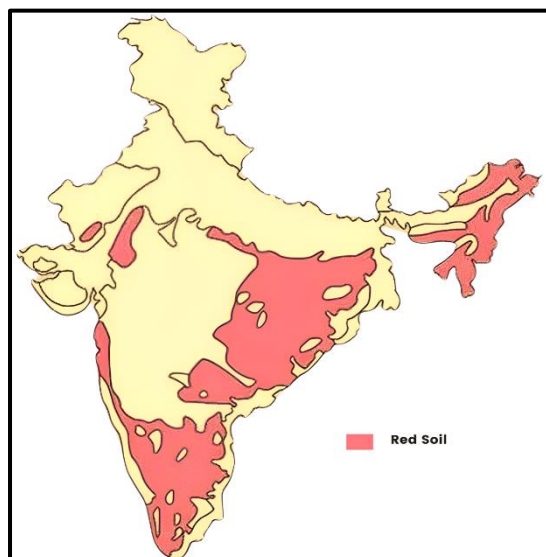


Distribution:

- Red soils are primarily found in the **Peninsular region** of India. Their distribution extends from **Tamil Nadu in the south to Bundelkhand in the north**, and from **Rajmahal in the east to Kathiawad in the west**. This extensive distribution highlights their importance in Indian agriculture and land use.

Agricultural Importance:

- Despite their natural nutrient deficiencies, red soils can be **highly productive when managed properly**. When irrigated and supplemented with humus and other nutrients, they can yield high outputs, particularly in crops such as rice, sugarcane, and cotton. *{ Dry crops }*



- In **drier areas**, red soils support the cultivation of **millet**s and **pulses**. The Kaveri and Vaigai basins, known for their red alluvial soils, are particularly suitable for **paddy cultivation** when adequately irrigated. Additionally, large regions in Karnataka and Kerala have utilized red soil areas for **rubber and coffee plantations**, demonstrating the versatility and agricultural potential of these soils.

Kerala mai hota to red soil

Laterite Soils:

Laterite soil derives its name from the **Latin word "later,"** meaning **'brick.'** This type of soil forms primarily in regions characterized by **high temperatures** and **heavy rainfall**, conditions typical of tropical climates. The tropical rains lead to **intense leaching**, a process that **removes silica and lime from the soil**, leaving behind **residues rich in iron oxides and aluminium compounds**. It covers about **3.7% of the country's total area**.

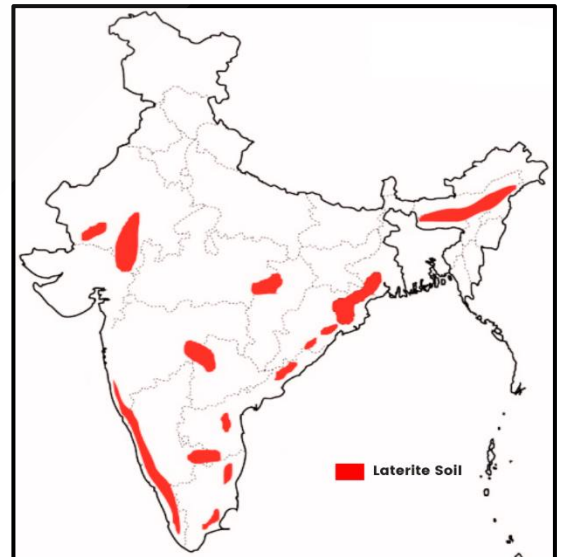
Characteristics:

- Laterite soil typically has a **brown or reddish-brown colour**, a result of its **high iron oxide** content, which often occurs in the form of nodules. The **iron oxides are often found in the form of nodules**, which contribute to the soil's distinct properties.
- Laterite soil is notably **low in organic matter, nitrogen, phosphate, and calcium**, which are essential nutrients for plant growth. This deficiency is partly due to the **rapid decomposition of organic matter**, as the **high temperatures encourage bacterial activity**, which quickly breaks down humus.
- Conversely, the soil is **rich in iron oxide and potash**. Due to these imbalances, laterite soil is generally **not suitable for cultivation unless supplemented with manures and fertilizers** to improve its fertility.



Distribution:

- Laterite soil is distributed in various regions across India, often occurring in patches. In the **Western Ghats**, it is found in the states of **Goa** and **Maharashtra**. The **Belgaum district of Karnataka** and the **laterite plateaus of Kerala** are also known for this soil type.
- Additionally, laterite soil is present in the state of **Odisha**, particularly in the **Eastern Ghats**, as well as in the **Amarkantak plateau region of Madhya Pradesh**. In **Gujarat**, it is found in the **Panchmahal district**, and in **Jharkhand**, it is common in the **Santhal Pargana divisions**.



Agricultural Importance:

- Although laterite soils are typically poor for general agricultural use, they **can be made productive with appropriate soil management practices**.
- In regions like **Tamil Nadu, Andhra Pradesh, Kerala, and Karnataka**, the red laterite soils have been found suitable for growing tree crops such as **groundnut, cashew nut**, and various **tree crops**, including **coffee, rubber, and spices**, **particularly in the laterite soils of**.
- Additionally, laterite soil is commonly used as a **building material**, where it is **cut into bricks** for **constructing houses**, especially in regions like **Kerala** and **Karnataka**.

Arid Soils (Desert Soils):

Arid soils are primarily found in **regions with dry climates**. These soils are associated with desert and semi-desert conditions characterized by **prolonged periods of water scarcity**, often lasting up to six months. Their formation is largely influenced by limited moisture availability, resulting in unique physical and chemical properties.

Characteristics:

- Desert soils typically have a **sandy to gravelly texture**, exhibiting **poor organic matter content** and **low levels of humus**. Due to infrequent rainfall and extended drought seasons, these soils show **poorly developed horizons**. The colour of desert soils ranges from **red to light brown**, reflecting the mineral composition and **iron content prevalent** in these areas. The **sparse vegetation** often seen in these regions is a result of wide spacing between plants, indicating the limited nutrient and moisture availability.
- One of the defining features of arid soils is the presence of a 'Kankar' layer in the **lower horizons**. This layer forms **due to an accumulation of calcium carbonate**, which increases in concentration with depth. The kankar layer **restricts the infiltration of water, creating a barrier that can affect water availability for plant growth**. However, when irrigation is applied, this layer can facilitate the retention of moisture, thereby supporting sustainable agricultural practices.
- Arid soils are typically **low in essential nutrients**, particularly **nitrogen**, which is vital for plant growth. Additionally, the **phosphate content in these soils is also insufficient**, further limiting their agricultural productivity. These nutrient deficiencies, combined with the challenging climatic conditions, make arid soils **less suitable for conventional farming practices without appropriate soil management and irrigation techniques**.

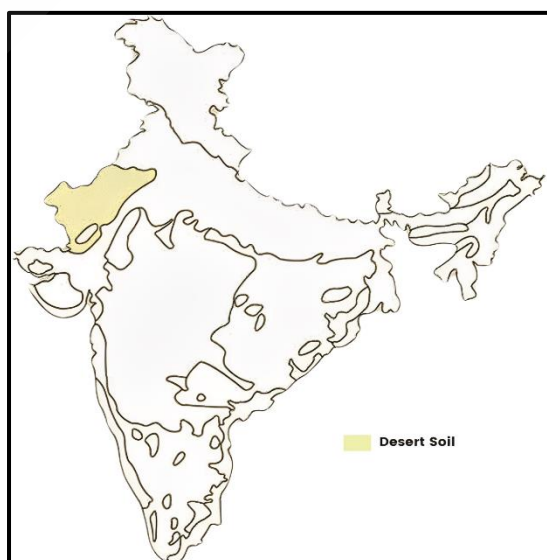


Distribution:

- Desert soils are predominantly distributed across **western Rajasthan, the Rann of Kachchh**, and in scattered patches in **southern Haryana and southern Punjab**. This distribution is a testament to the climatic and geographical factors that define these arid and semi-arid landscapes.

Agricultural Importance:

- Despite their harsh characteristics, desert soils have the **potential for agriculture if adequate water supply is provided**. With proper irrigation and the **application of fertilizers**, these soils can support a variety of crops.
- Less water-intensive crops** such as **bajra** (pearl millet), **pulses, fodder, and guar** (cluster bean) thrive in these environments. When irrigation is implemented, desert soils can yield high agricultural returns, showcasing their potential for productive land use.

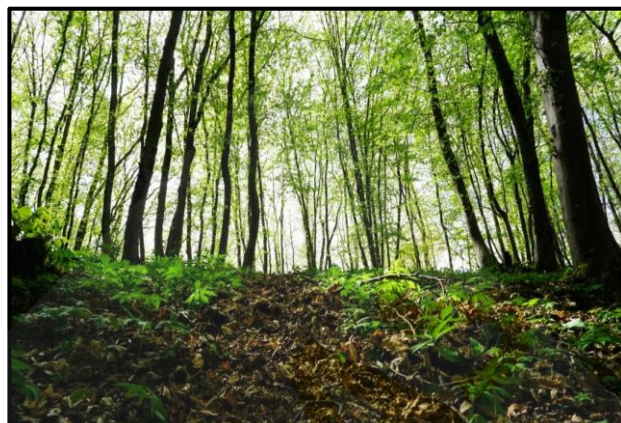


Forest Soils (Mountain Soils):

Forest or mountain soils are predominantly found in **mountainous regions** characterized by **steep slopes, high relief, and shallow soil profiles**. The challenging terrain and elevation influence the soil's development, resulting in a thin, poorly developed profile with distinct horizons. The rapid drainage in these areas makes the soil susceptible to erosion, especially on steeper slopes.

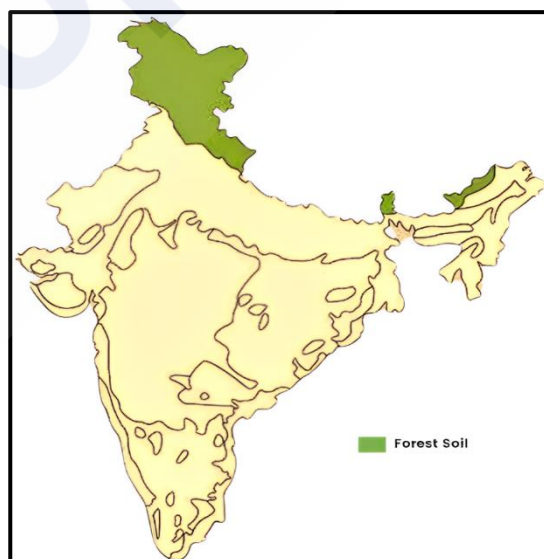
Characteristics:

- The characteristics of these soils vary significantly depending on the specific mountain environment in which they develop. On **valley sides, forest soils tend to be loamy and silty**, which provides a fertile ground for vegetation. However, on the **upper slopes, these soils are generally coarse-grained** due to the harsher environmental conditions.
- In the **snow-bound regions** of the Himalayas, forest soils **undergo significant denudation, which affects their composition**. These soils are **often acidic and have low humus content, making them less fertile compared to other areas**. On the other hand, the **soils found in the lower valleys are typically more fertile and supportive of plant growth**.
- Mountain soils are typically **thin-layered with poorly developed profiles** and horizons due to the **rapid drainage and continuous erosion**.



Distribution:

- Mountain soils are generally found at **altitudes above 900 meters**, particularly in regions such as the **Himalayas, the foothills of the Himalayas, and the mountainous slopes of the Western Ghats, Nilgiri Hills, Annamalai Hills, and Cardamom Hills**. These soils are widely distributed across various elevations and climates, contributing to their complexity and variability.



Agricultural Importance:

- The unique properties of mountain soils make them ideal for crops that require good air and water drainage, which are naturally provided by the slopes. These soils are **extensively used for rubber plantations, bamboo cultivation, as well as tea, coffee, and fruit farming**. However, the fertility of mountain soils tends to decline after two to three years of cultivation, particularly in areas where shifting agriculture is practiced.
- Given the **limited potential for extensive agriculture** in these regions, **silvopastoral farming—an integrated approach combining forestry and grassland—can be a sustainable alternative**. Different types of crops are cultivated in various regions depending on the altitude and soil type. For example, rice is typically grown in the valleys, orchards are established on the slopes, and potatoes are cultivated in almost all areas.

Saline and Alkaline Soils:

Saline and alkaline soils are characterized by a **high concentration of soluble salts**, including **sodium, potassium, and magnesium**. This high salt content renders the **soils infertile**, significantly limiting their ability to support vegetation. These soils are commonly referred to by various local names such as **Reh, Usar, Kallar, Rakar, Thur, and Chopan**. The prevalence of salts in these soils is primarily due to factors such as dry climatic conditions and inadequate drainage, making them common in arid and semi-arid regions, as well as in areas that are waterlogged or swampy.

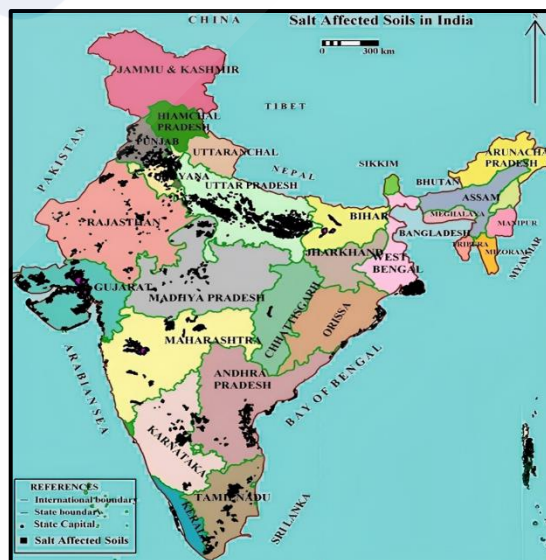
Characteristics:

- The texture of saline soils can vary from **sandy to loamy**, but they are generally **deficient in essential nutrients like nitrogen and calcium**. The **excessive presence of salts**, such as **sodium chloride** and **sodium sulphate**, makes them unfavourable for most agricultural activities.



Formation and Distribution:

- The formation of saline and alkaline soils can be attributed to both natural and anthropogenic factors.
 - Natural Formation** - These soils naturally occur in areas such as the **dried-up lakes of Rajasthan and the Rann of Kutch**. The Palaya basin, a clay basin located in the midst of the desert, is another region where these soils have emerged. The **evaporation of water in these areas leaves behind high concentrations of salts**, contributing to the formation of saline and alkaline soils.
 - Anthropogenic Formation** - Human activities, particularly faulty **agricultural practices**, have led to the development of these soils in regions like **western Uttar Pradesh and Punjab**. **Improper irrigation methods, excessive use of water, and inadequate drainage systems** contribute to the accumulation of salts in the soil, exacerbating its salinity and alkalinity.
 - These soils are predominantly found in regions like **western Gujarat, the deltas along the eastern coast, and the Sundarbans in West Bengal**.



Agricultural Importance:

- To mitigate soil salinity, particularly in **intensively cultivated areas**, farmers are often advised to **add gypsum to the soil**. Gypsum helps to **displace sodium ions from the soil, replacing them with calcium**, which **reduces the soil's salinity and restores its fertility**, making it more suitable for agriculture.
- With proper management and support, they can support the growth of **leguminous crops**, which are more tolerant of saline conditions. Despite this, the overall fertility of these soils is significantly reduced, making them less suitable for conventional farming.

Peaty and Marshy Soil (Bog Soil):

Peaty and marshy soil, also known as **bog soil**, forms in areas where **proper drainage is not feasible**, leading to **waterlogged conditions**. These soils are **rich in organic matter**, often composed of **partially decomposed plant material**, which accumulates over time in the **absence of adequate oxygen**. Due to the poor drainage and the presence of stagnant water, these soils are also characterized by **high salinity levels**.

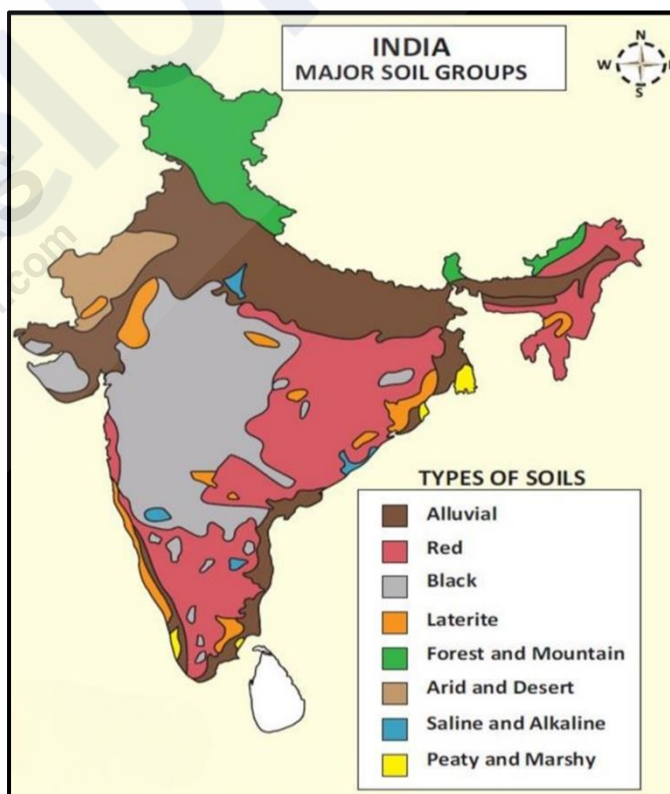
Characteristics:

- Peaty and marshy soils are primarily composed of **clay and mud**, which contribute to their heavy texture. The **high moisture content** in these soils makes them particularly water-retentive, but the **excessive salt content, combined with frequent inundation by high tides**, renders them **largely infertile**.
- The persistent moisture **inhibits organic activity**, as the waterlogged conditions create an **anaerobic environment** that **limits the decomposition of organic material**, further reducing soil fertility.
- They are notably **deficient in essential nutrients like potash and phosphate**, which are crucial for plant growth.



Distribution:

- This type of soil is predominantly found in the **delta regions of India**, where river systems deposit large amounts of silt and clay in low-lying areas. Notably, it is characteristic of the **Bengal Delta (Sundarbans)**, where the soil supports the cultivation of crops like **jute and rice**.
- Beyond the delta regions, peaty and marshy soils are also present in specific areas such as **Alleppey in Kerala**, where they are referred to as "**Karri**" along the backwaters or "**Kayals**." Additionally, these soils can be found in **Almora, Uttarakhand**, where similar conditions of poor drainage and high organic content prevail.



Agricultural Importance:

- Despite their general infertility, peaty and marshy soils hold some agricultural significance. In the Bengal Delta, the soil is suitable for the cultivation of **jute and rice**, two important crops in the region.
- In the Malabar region of Kerala, the soil, with appropriate management, can support the growth of **spices, rubber, and large-sized varieties of rice**. Additionally, these soils play a role in supporting the unique ecosystems of mangrove forests, particularly in coastal areas, where they contribute to the stability and health of these vital habitats.

Soil Problems:

Soil Erosion:

Soil erosion is the process where the **top fertile layer of soil is removed by natural forces like wind, water, or human activity**. It is a major environmental concern as it leads to the loss of soil fertility, reduces agricultural productivity, and can lead to sedimentation in water bodies, which affects aquatic life.

Types of Soil Erosion:

- **Water Erosion** - Caused by rainfall and surface runoff, leading to rill, gully, and sheet erosion.
- **Wind Erosion** - Common in arid and semi-arid regions, where strong winds blow away loose, dry, and fine soil particles.
- **Glacial Erosion** - Occurs in cold regions where glaciers move, dragging soil and rocks with them.
- **Human-Induced Erosion** - Activities like deforestation, overgrazing, and improper agricultural practices accelerate soil erosion.

Impact:

- Loss of topsoil and nutrients.
- Decreased agricultural yield.
- Sedimentation in rivers and lakes, affecting water quality.
- Desertification in extreme cases.

Soil Degradation:

Soil degradation refers to the **decline in soil quality caused by various factors, including excessive agricultural practices, deforestation, overgrazing, industrial activities, and urbanization**. It involves the physical, chemical, and biological deterioration of soil.

Types of Soil Degradation:

- **Physical Degradation** - Includes compaction, waterlogging, and soil crusting, which reduce soil permeability and root growth.
- **Chemical Degradation** - Involves the loss of soil nutrients, acidification, salinization, and contamination by heavy metals or chemicals.
- **Biological Degradation** - A decline in the population of beneficial soil organisms and a reduction in organic matter content.

Impact:

- Reduced soil fertility.
- Lower crop yields.
- Increased vulnerability to erosion.
- Loss of biodiversity.

Soil Pollution:

Soil pollution occurs when **harmful substances like chemicals, heavy metals, or waste materials are introduced into the soil, contaminating it and making it unfit for agriculture or other uses.**

Sources of Soil Pollution:

- **Agricultural Chemicals** - Overuse of pesticides, herbicides, and fertilizers.
- **Industrial Waste** - Discharge of toxic substances like heavy metals and chemicals from factories.
- **Urban Waste** - Disposal of untreated sewage, plastics, and other non-biodegradable waste.
- **Mining Activities** - Release of heavy metals and other pollutants during extraction processes.

Impact:

- Contamination of food crops.
- Health hazards to humans and animals.
- Disruption of soil ecosystems.
- Decreased soil fertility.

Wrong Agricultural Practices:

Certain **agricultural practices can lead to soil degradation** and erosion. These include:

- **Monoculture** - Growing the same crop year after year depletes specific nutrients from the soil, leading to reduced fertility.
- **Overgrazing** - Excessive grazing by livestock removes vegetation cover, exposing soil to erosion.
- **Over-irrigation** - Leads to waterlogging and salinization, especially in poorly drained soils.
- **Improper Tillage** - Frequent and deep plowing can disturb soil structure, leading to erosion and compaction.

Impact:

- Nutrient depletion.
- Increased erosion risk.
- Soil compaction.
- Lower agricultural productivity.

Soil Desertification:

Soil desertification is the process by which **fertile land becomes desert due to factors such as drought, deforestation, and inappropriate agriculture.** It is characterized by a severe decline in soil fertility and the loss of vegetation cover.

Causes of Desertification:

- **Climate Change** - Prolonged droughts and reduced rainfall.
- **Deforestation** - Removal of trees that anchor the soil.
- **Overgrazing** - Excessive livestock grazing leading to loss of vegetation.
- **Unsustainable Farming** - Practices that deplete soil nutrients and water resources.

Impact:

- Loss of arable land.
- Reduced agricultural output.
- Migration and displacement of populations.
- Ecological imbalance.

Solutions to Soil Problems (Soil Conservation):

Soil conservation involves a range of **practices aimed at preventing soil erosion, degradation, and pollution**. These practices help maintain soil health, ensuring its long-term productivity and ecological function.

- **Contour Plowing** - Contour plowing involves plowing along the contours of the land rather than up and down slopes. This method helps reduce surface runoff and soil erosion by allowing water to infiltrate the soil rather than washing it away.
- **Terracing** - Terracing is the creation of stepped levels on sloping land. It reduces runoff and soil erosion by breaking the slope into smaller, more manageable sections, each acting as a barrier to slow down water flow.
- **Crop Rotation and Diversification** - Crop rotation involves alternating the types of crops grown in a particular field from one season to the next. This practice helps maintain soil fertility and reduces the risk of soil-borne diseases. Diversifying crops also helps in improving soil structure and organic matter content.
- **Afforestation and Reforestation** - Planting trees and maintaining forest cover helps protect the soil from erosion, improves water retention, and enhances biodiversity. Trees act as windbreaks, reducing wind erosion, and their roots help bind the soil, preventing it from being washed away by water.
- **Use of Organic Fertilizers** - Organic fertilizers, such as compost and manure, improve soil structure, enhance nutrient content, and increase the population of beneficial soil organisms. Unlike chemical fertilizers, they do not cause soil pollution or degradation.
- **Controlled Grazing** - Implementing rotational grazing and limiting the number of livestock per unit area can prevent overgrazing, allowing vegetation to recover and reducing soil erosion.
- **Mulching** - Mulching involves covering the soil with organic or inorganic materials like straw, leaves, or plastic. This practice helps retain soil moisture, reduce erosion, and suppress weed growth.
- **Conservation Tillage** - Conservation tillage reduces the frequency and intensity of tilling, leaving crop residues on the field. This practice helps maintain soil structure, reduces erosion, and increases water infiltration.
- **Soil Reclamation** - In areas affected by salinization or acidification, soil reclamation techniques, such as leaching salts with water, applying gypsum, or liming acidic soils, can restore soil productivity.
- **Pollution Control** - Preventing soil pollution involves proper waste management, reducing the use of harmful chemicals in agriculture, and controlling industrial discharges. Remediation techniques like phytoremediation, where plants absorb contaminants, can also be used to clean up polluted soils.

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GS FOUNDATION 2024-25

GEOGRAPHY - 23

Physical Setting of India - I

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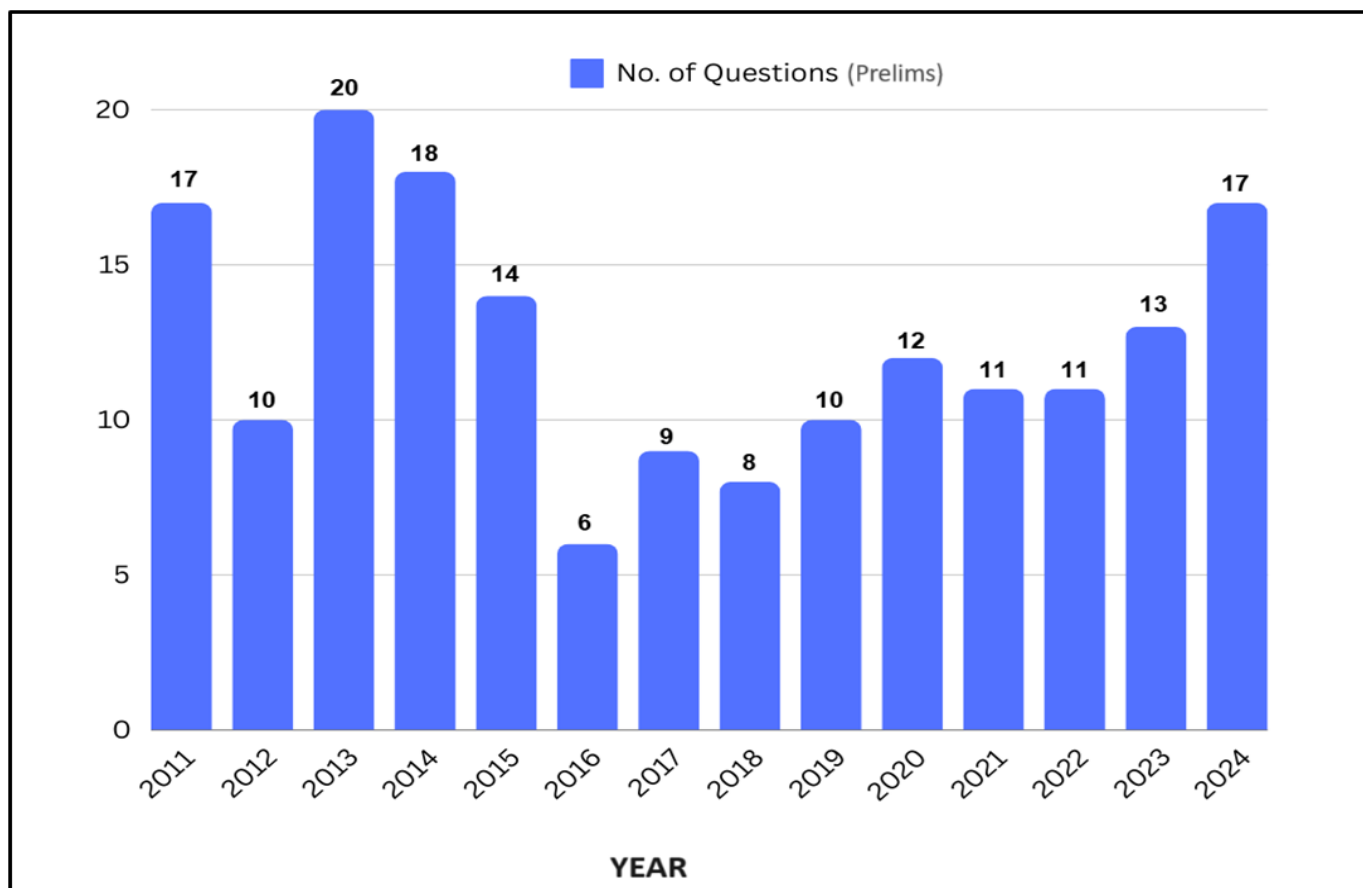
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Year-wise Trend Analysis (2011 – 2024)



Topic-wise Trend Analysis (2011 – 2024)

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Size and Location

Introduction:

India is a country of immense geographical diversity, stretching from the icy **peaks of the Himalayas in the north** to the **tropical shores of the Indian Ocean in the south**. Located in Asia, the largest continent in the world, India forms a significant part of South Asia. It is geographically separated from the rest of the continent by the towering Himalayas, which contribute to the country's unique climatic and topographical features. India's vast landmass is home to a **variety of landscapes, ranging from the snow-covered heights of the northern Himalayas to the fertile plains of the Indo-Gangetic region, and the arid expanses of the Thar Desert in the west**. The southern part of the country is dominated by the Peninsular Plateau, which is flanked by the Eastern and Western Coastal Plains. The tropical sun and monsoonal rains play a crucial role in shaping the climate and life in India, making these two climatic factors vital for the country's people. Due to its immense size and diversity, **India is often referred to as a subcontinent**, possessing many characteristics typically associated with a continent.

Nomenclature of India:

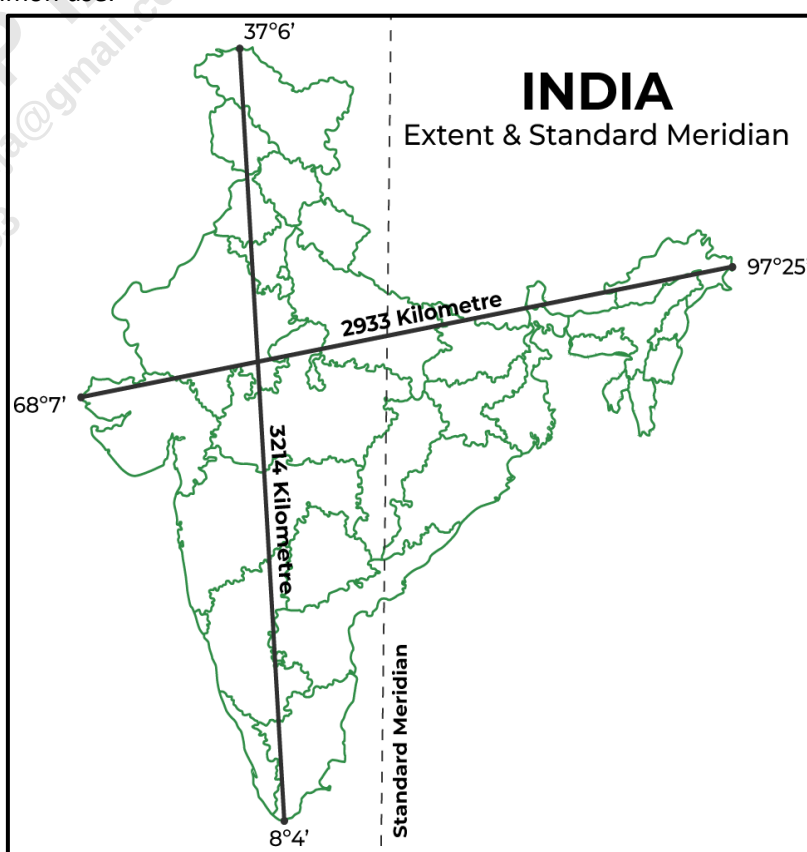
The country has been known by various names throughout history, including **India, Hindustan, Bharat, and Aryavarta**. The name "**India**" originates from Greek literature, referring to the land of the "**Indoi**," or **the people living near the Indus River**. The Persians and Greeks extended the name "**Sindhu**," the river, to the entire region, which came to be known as **Hindustan** in Persian and other West Asian languages. The term "Hindu" is derived from "**Sindhu**," with the Persians pronouncing "S" as "H," leading to the name "**Hindu**" for the land to the east of the Sindhu River.

In Hindu literature, the subcontinent is referred to as **Bharat or Bharat-Varsha**, named after the legendary **King Bharata** who is said to have visualized the fundamental unity of the country. Some scholars believe the name "**Bharat**" is derived from the **Bharath tribe** that inhabited the region. In European languages, the country is popularly known as India, while "**Aryavarta**" refers to the land of the Aryan race. Today, both "**India**" and "**Bharat**" are officially recognized names of the country, although "**Hindustan**" remains in common use.

Location of India:

India is **entirely located in the Northern Hemisphere**, with the Indian mainland **extending from 8°4'N to 37°6'N latitudes** and from **68°7'E to 97°25'E longitudes**. This results in a latitudinal and longitudinal extent of about **29 degrees**.

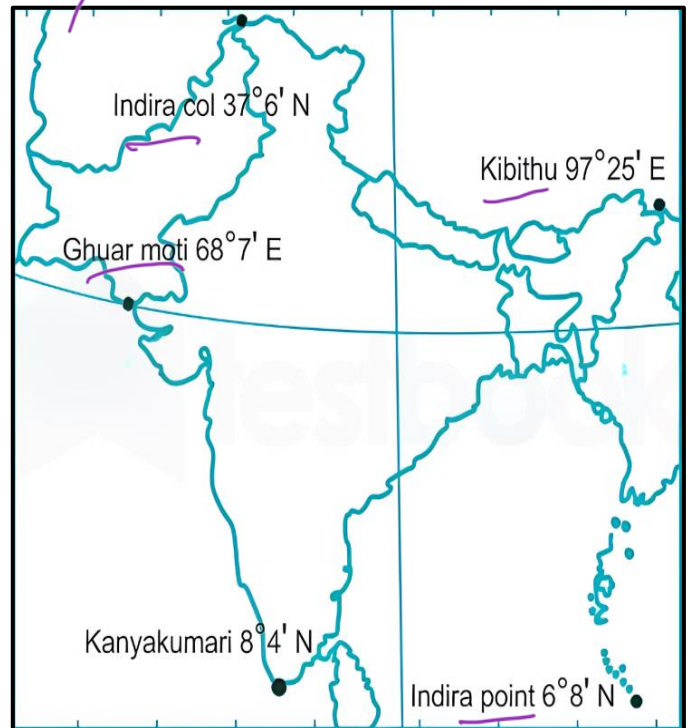
The country measures **approximately 3,214 kilometers from north to south** and **2,933 kilometers from east to west**. However, despite the similar extent in degrees, the actual distances differ considerably due to the **curvature of the Earth**. The east-west distance between two successive meridians of longitude along the equator is at its maximum—about 111 kilometers—and gradually decreases as one



moves toward the poles, where the distance becomes zero. This is because all meridians converge at the poles. In contrast, the north-south distance between any two successive parallels of latitude along a meridian remains relatively uniform at **approximately 111 kilometers**.

The **northernmost point of the Indian mainland** is located in **Jammu and Kashmir (Indira Col)**, while the **southernmost point is Kanyakumari in Tamil Nadu**. However, the southernmost point of the entire country is **Indira Point** in the Andaman and Nicobar Islands, situated at **6°30'N latitude**.

The **westernmost point of India** is in **Gujarat (Ghuar Moti)**, and the **easternmost point is in Arunachal Pradesh (Kibithu)**.



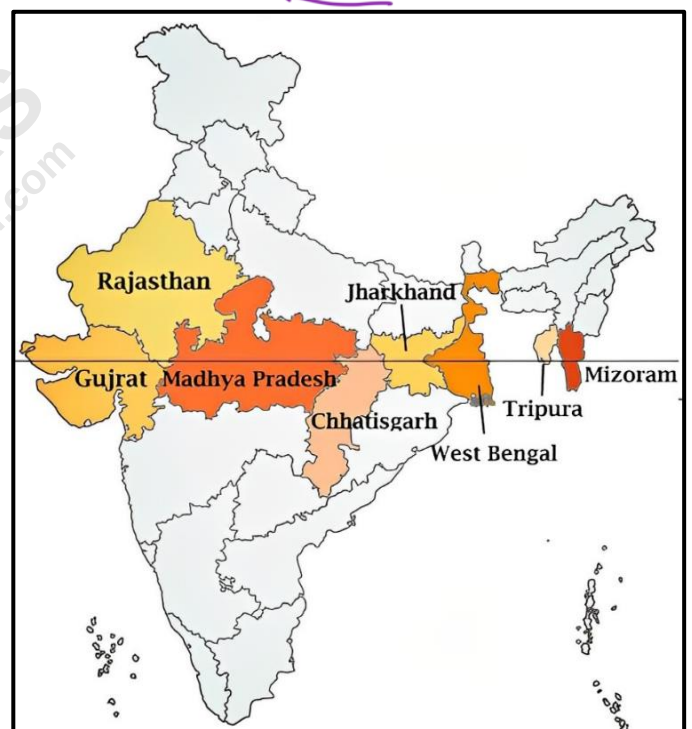
India's large latitudinal extent significantly affects the climate and the length of day and night across the country. The northern parts of India, being farther from the equator, receive the Sun's rays more obliquely, resulting in less insolation and a colder climate compared to the southern parts.

Additionally, the **difference between the length of day and night** varies considerably across the country. In the **southernmost parts of India, near the equator, the difference is minimal—about 45 minutes—**while in the **northern parts, it can be as much as 5 hours**.

Tropical and Subtropical Zones:

The **Tropic of Cancer** passes almost midway through the country, dividing it into **two climatic zones**. The southern half of India, lying south of the Tropic of Cancer, is situated in the **Tropical or Torrid Zone**, characterized by a warm climate. The northern half, lying north of the Tropic of Cancer, falls in the **Subtropical Zone**, which experiences a relatively cooler climate.

The Tropic of Cancer passes through **8 states in India** - **Gujarat** (Jasdan), **Rajasthan** (Kalinjarh), **Madhya Pradesh** (Shajapur), **Chhattisgarh** (Sonhat), **Jharkhand** (Lohardaga), **West-Bengal** (Krishnanagar), **Tripura** (Udaipur) and **Mizoram** (Champhai).



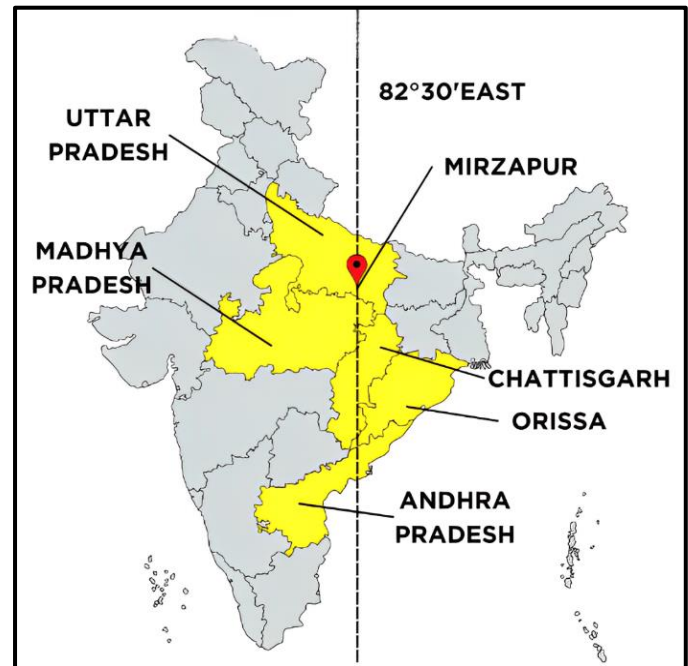
Peninsular Shape and Maritime Influence:

India's geographical shape significantly influences its maritime boundaries. The northern part of India is broad, while the **southern part tapers towards the Indian Ocean**, effectively dividing the northern Indian Ocean into two distinct parts: the **Arabian Sea to the west** and the **Bay of Bengal to the east**. The total length of India's coastline, including the island groups, is approximately **7,516.6 kilometers**. The **Palk Strait**, a narrow body of water, separates the Indian mainland from Sri Lanka. Structurally, Sri Lanka is considered an extension of the Indian peninsular block.

Time Zone and Standard Time:

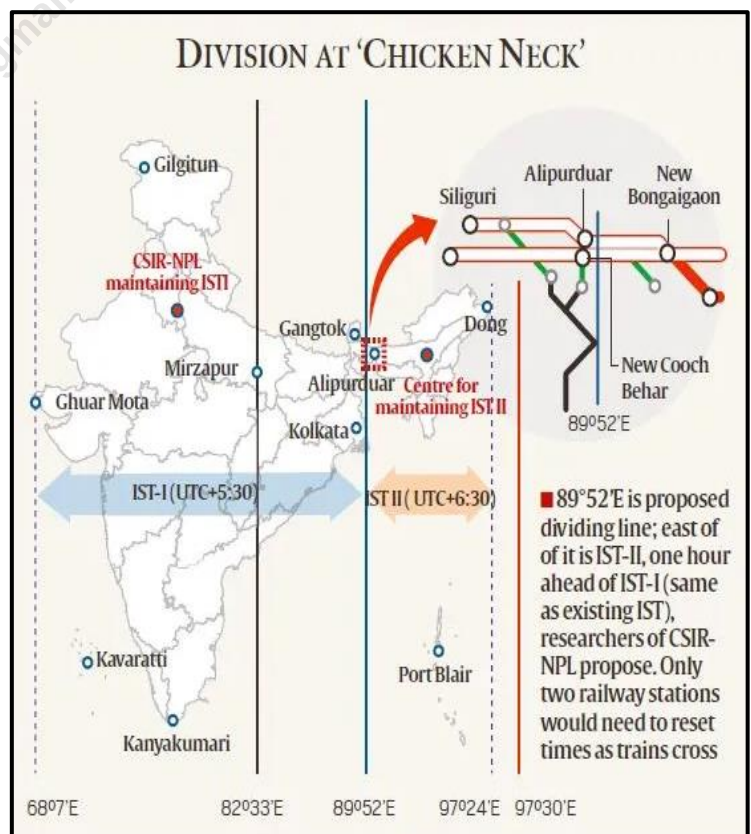
Due to India's **longitudinal extent of nearly 29°**, there is a **real-time difference of about two hours** between its eastern and western extremities. While the eastern extremity experiences sunrise, the western extremity would still have nearly two hours before the day begins. To manage this time difference, India, like other countries, follows a single standard time based on a central meridian.

The standard meridian of India is 82°30'E, which has been chosen as it lies approximately in the middle of the country's longitudinal extent. This ensures uniform time across the nation. India's Standard Meridian (IST) passes through **5 states in India - Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Odisha, and Andhra Pradesh**.



Need for Two Time Zones:

- **Large Geographical Extent** - India is the second-largest country geographically, after China, that operates on a single time zone. The country spans from **97°25'E in Arunachal Pradesh to 68°7'E in Gujarat**, covering nearly **30 degrees of longitude**, which is sufficient to justify the implementation of two time zones.
- **Early Sunrise and Sunset** - The call for two time zones has gained traction due to the early sunrise and sunset experienced in northeastern India (NER) and the Andaman and Nicobar Islands (ANI). In these regions, the **sun rises and sets much earlier compared to the rest of the country**. The NER has long raised concerns about the impact of a single time zone on daily life and economic activities, as valuable daylight hours are lost. In summer, the sun can rise as early as 4 am, but government offices do not open until 10 am, resulting in inefficiencies.
- **Sync with Circadian Rhythm** - Many Indians are living and working in a time zone that does not align with their natural diurnal cycles. **Productivity and efficiency** are closely tied to the biological clock, which is synchronized with the light-dark cycle. From a circadian rhythm perspective, the current Indian Standard Time (IST) suits Kanyakumari, Kavaratti, and Ghuar Mota well, is manageable for places like Alipurduar, Kolkata, Gangtok, Mirzapur, and Gilgitum, but is highly unsuitable for locations like Dong and Port Blair.
- **CSIR's Suggestions** - In **2006**, the **Planning Commission recommended** splitting the country into two time zones. The **CSIR-National Physical Laboratory (NPL) has endorsed** this long-standing demand, proposing two ISTs separated by a one-hour difference:
 - **IST-I** for most of India, covering regions between **68°7'E and 89°52'E**.
 - **IST-II** for the Northeastern region, covering areas between **89°52'E and 97°25'E**.



Advantages of Having Two Time Zones:

- **Efficiency** - Adopting two time zones could **enhance workforce efficiency and optimize energy consumption**. People would be better able to work and plan according to their natural cycles. For instance, tea gardens in Assam have long operated on an informal time zone known as "**Chaibagaan Time**," which is set one hour ahead of IST.
- **Environmental Protection** - Advancing IST by just half an hour could **save 2.7 billion units of electricity annually** by utilizing the wasted daylight hours. This conservation is crucial for India's economy, which faces a significant power deficit. According to CSIR-NPL, India could **save ₹1,000 crore per year** by conserving electricity through the implementation of two time zones. Reducing energy consumption would also significantly lower India's carbon footprint, supporting its commitment to combating climate change.
- **Societal Benefits** - Two time zones could help **achieve various social policy objectives, such as reducing road accidents and improving women's safety**. Research indicates that people in countries with multiple time zones can better focus on their strengths and weaknesses, leading to improved outcomes. Adjusting the time zone also allowed employees to **develop healthier eating, sleeping, and working habits, yielding positive results**.

Disadvantages of Having Two Time Zones:

- **Timing Mismatch** - The introduction of two time zones could lead to **mismatches in office timings, varying working hours for banks, and potentially more frequent railway accidents**. Implementing two time zones would require careful synchronization of railway traffic to avoid confusion. It could also disrupt travel schedules, as most people travel by road, not air, unlike in the West where adjusting clocks suffices.
- **Chaos at Border of Time Zones** - Given India's significant **illiteracy levels and low awareness**, dividing the country into two time zones could create chaos at the borders of these zones. People would need to reset their clocks each time they cross a time zone boundary. Additionally, marking the dividing line between the two zones would pose a challenge.
- **Prone to Increase Divide** - Implementing two time zones could have **adverse political consequences, potentially exacerbating divisions in India**, which is already divided along lines of religion, caste, race, and language. A separate time zone for the NER could foster a sense of separation from the rest of the country, possibly fueling secessionist sentiments.
- **Infrastructural Burden** - Establishing two time zones would necessitate a second laboratory for CSIR-NPL in the new time zone. This would require setting up a '**Primary Time Ensemble-II**,' traceable to the **Coordinated Universal Time (UTC)** at the International Bureau of Weights and Measures (BIPM) in France.

Chaibagaan Time

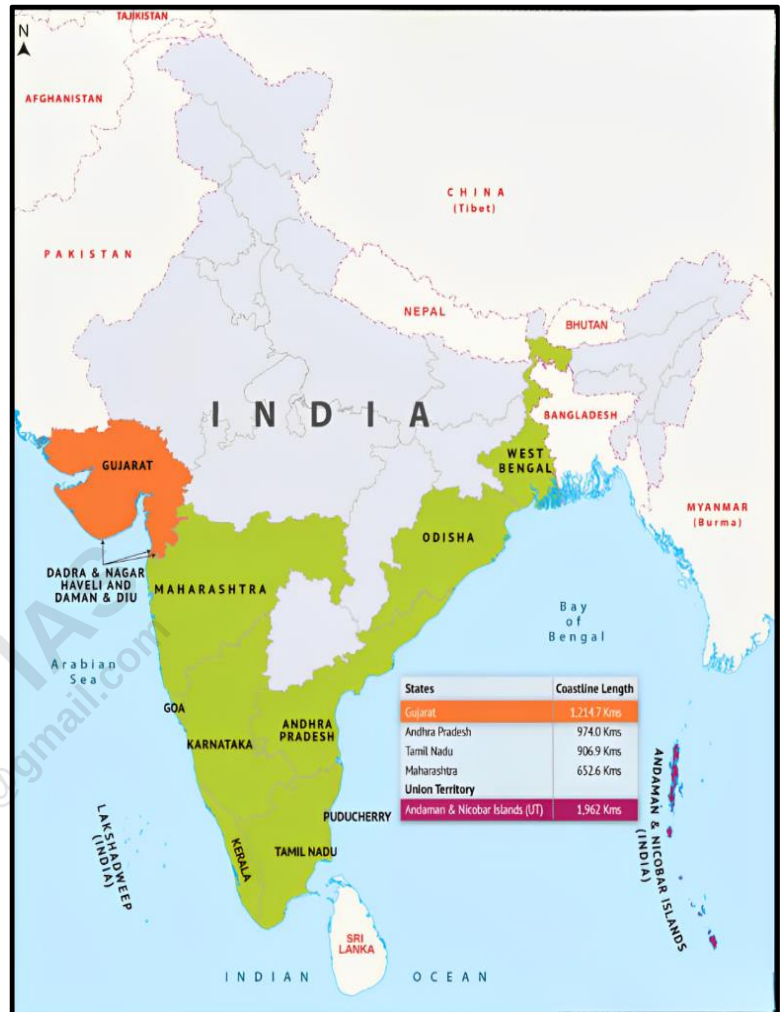
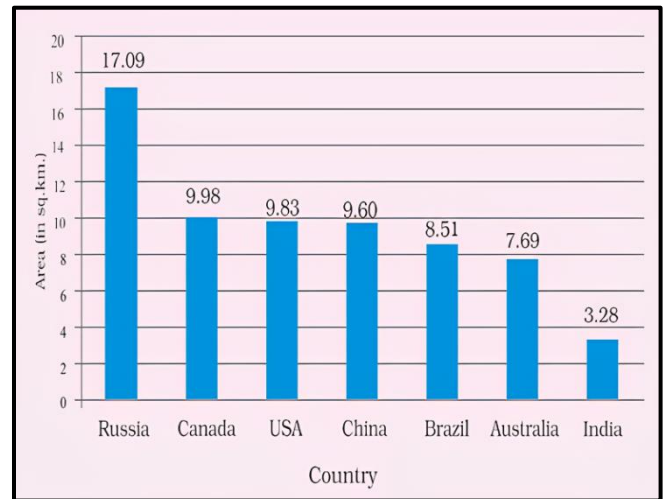
*150 years ago, British colonialists introduced "Chaibagaan time" or "Bagaan time", a time schedule observed by tea planters, which was **one hour ahead of IST**. This was done **to improve productivity by optimizing the usage of daytime**. After Independence, Assam, along with the rest of India, has been following IST for the past 66 years. The administration of the Indian state of Assam now wants to change its time zone back to Chaibagaan time to conserve energy and improve productivity. Indian government didn't accept to such a proposal.*

Size of India:

India stands as the **seventh-largest country** in the world by land area, spanning an extensive **3.28 million square kilometres**. The nation is marked by a significant land boundary of **15,200 km** and a coastal boundary, including its islands, that stretches to 7,516 km. Despite occupying a mere **2.4% of the world's total land area**, India's size and geographical diversity are remarkable.

On the map of India, prominent natural features like the **youthful fold mountains** dominate the landscape, particularly in the **north, northwest, and northeast**, forming **natural boundaries**.

India's coastline extends for 7,516.6 kilometers, including the **mainland coastline of 6,100 kilometers** and the **coastline of Indian islands**, which measures **1,197 kilometers**. This extensive coastal stretch runs through **13 states and union territories of the country**. The formation of India's coastline is linked to the tectonic activities associated with the faulting of Gondwanaland during the Cretaceous period. This geological event, driven by the process known as Continental Drift, resulted in the creation of a straight and regular coastline. However, due to its geological origins, India's coast has a **limited number of naturally favourable harbours**. The drifting and separation of landmasses following the breakup of Gondwanaland during the Cretaceous or early Tertiary periods also played a crucial role in shaping the Indian subcontinent's coastal features, giving rise to the Bay of Bengal and the Arabian Sea.



Significance of India's Location:

India's strategic geographical location has played a pivotal role in shaping its **diplomatic and economic relationships** with other nations. As a subcontinent, India shares **land borders with multiple countries** to the north and south, enhancing its connectivity with diverse regions. The southern half of India extends into the vast Indian Ocean, a factor of significant strategic importance for international trade. The country's **extensive coastline and well-established maritime channels** have facilitated active interaction with the global community. **The naming of the Indian Ocean after India underscores the country's central position within this vital water body.**

Historically, before the comprehensive exploration of sea routes, mountain passes served as crucial pathways for traders and travellers to enter and exit India, highlighting the enduring significance of India's geographical location in facilitating cross-border movement and trade. **India's central positioning in Asia has been a catalyst for fostering trade and promoting its active participation in global affairs.** This advantageous geographic placement has been instrumental in enhancing India's economic and diplomatic engagement with the broader international community.

India and its Neighbours:

Most of India's states have either international borders or coastlines, making them critical from a border management perspective. States like **Madhya Pradesh, Chhattisgarh, Jharkhand, Delhi, Haryana, and Telangana** do not share **international borders or coastlines**. In contrast, other states have one or more international borders or coastlines, categorizing them as frontline states in terms of border security. The country has a land border stretching over **15,106.7 km**, running through **92 districts** across **17 states**.

India shares its **longest land border** with **Bangladesh**, extending over **4,096.7 km**, while its **shortest border**, measuring **106 km**, is with **Afghanistan**.

Name of the country	Length of the border (in Km)
Bangladesh	4,096.7
China	3,488
Pakistan	3,323
Nepal	1,751
Myanmar	1,643
Bhutan	699
Afghanistan	106
Total	15,106.7

Border with China:

India shares its **second-longest border with China**, covering a distance of **3,488 km** running along **Arunachal Pradesh, Sikkim, Uttarakhand, Himachal Pradesh** and **Union Territory of Ladakh**. The Sino-Indian border is commonly divided into **three sectors** - the **Western sector**, the **Middle sector**, and the **Eastern sector**.

The Western Sector:

- India and China dispute the **Aksai Chin** region, with India claiming it as part of Ladakh (erstwhile Jammu and Kashmir) and China considering it part of Xinjiang. This dispute originated from the ambiguity in the **British demarcation of the border**, with the **Johnson Line of 1865** placing Aksai Chin under Indian control and the **McDonald Line of 1893** placing it under **Chinese control**. Currently, the **Line of Actual Control (LAC)** separates Indian-administered Ladakh from Aksai Chin, aligning with China's claim line.



The Middle Sector:

- The Middle sector sees **fewer disagreements**, with the boundary running along the watershed between **Himachal Pradesh** and **Uttarakhand** on the Indian side and Tibet on the Chinese side.

The Eastern Sector:

- The Eastern sector dispute centers on the **McMahon Line**, with China rejecting it as an illegal boundary. The dispute is rooted in the **1914 Shimla Convention**, where the Tibetan representatives, according to China, lacked the authority to sign the agreement delineating the McMahon Line.

Border with Nepal:

India shares a **1,751 km** long border with Nepal, touching the Indian states of **Uttarakhand, Uttar Pradesh, Bihar, West Bengal, and Sikkim**. India and Nepal share an **open border, allowing unrestricted movement of goods and people between the two countries**. The boundary largely runs along the foothills of the **Shiwalik Range**.

Disputed Territories:

- **Kalapani** - Kalapani, a valley administered by India as part of Uttarakhand, is situated on the route to **Kailash Mansarovar**. The dispute arises from differing interpretations of the source of the **Kali River**, which demarcates the boundary between India and Nepal as per the **1816 Treaty of Sugauli**.
- **Susta** - Susta, located on the bank of the **Gandak River**, is another disputed area between India and Nepal. The changing course of the Gandak River has led to the shifting of the Susta region, creating a boundary dispute.

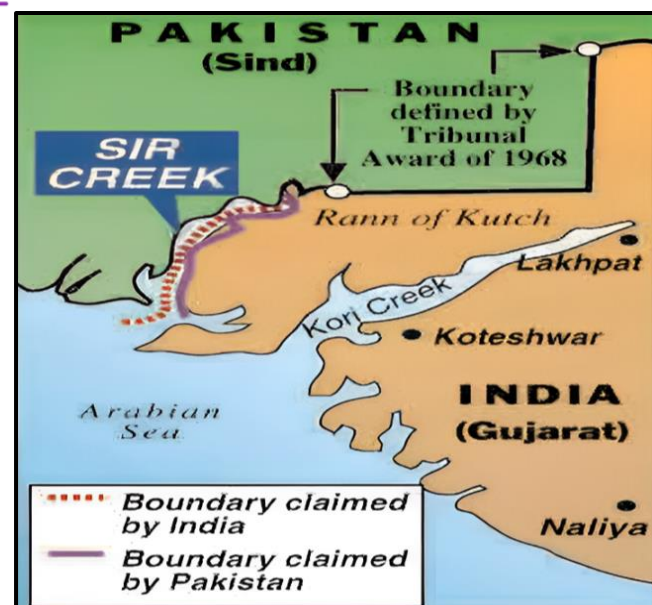


Border with Pakistan:

The Indo-Pakistan boundary, extending **3,323 km** running along **Gujarat, Rajasthan, Punjab, Union Territory of Jammu & Kashmir** and **Union Territory of Ladakh**. Resulted from the partition of India in 1947 under the **Radcliffe Award**, chaired by **Sir Cyril Radcliffe**.

Disputed Territories:

- **Jammu and Kashmir** - Pakistan occupies approximately 78,000 square kilometers of Indian territory in Jammu and Kashmir. Additionally, under the **1963 Sino-Pakistan Boundary Agreement**, Pakistan illegally ceded 5,180 square kilometers of territory in Pakistan-occupied Kashmir to China.
- **Siachen Glacier** - Located in the eastern **Karakoram range** of the Himalayas, the Siachen Glacier has been under **Indian administration since 1984**. The glacier is a highly strategic area, with ongoing military presence from both India and Pakistan.
- **Saltoro Ridge** - The Saltoro Ridge is a subrange of the **Karakoram**, located southwest of the Siachen Glacier. India controls the main peaks and passes, with Pakistani forces occupying the glacial valleys to the west.
- **Sir Creek** - Sir Creek is a **96 km long disputed estuary in the Rann of Kutch**. India and Pakistan differ in their interpretations of the boundary line, with India claiming the centerline and Pakistan claiming the eastern shore as the boundary.



Border with Bangladesh:

India shares its **longest border, 4,096.7 kilometers**, with Bangladesh. This boundary, established under the **Radcliffe Award**, resulted from the partition of the Bengal province.

Border with Myanmar:

The India-Myanmar boundary, spanning **1,643 kilometers**, runs roughly along the watershed between the **Brahmaputra and Ayeyarwady (Irrawaddy) rivers**. The boundary passes through thickly forested regions, including the Mizo Hills, Manipur, and Nagaland on the Indian side, and the Chin Hills, Naga Hills, and Kachin State on the Myanmar side.



Border with Sri Lanka:

India and Sri Lanka are separated by the narrow **Palk Strait**. The closest points are **Dhanushkodi** on the Tamil Nadu coast in India and **Talaimanar** in the **Jaffna peninsula** in Sri Lanka, connected by a group of islets forming **Adam's Bridge**. While the Indo-Sri Lanka boundary is generally peaceful, there has been tension over the ownership of **Kachchatheevu Island**, which India ceded to Sri Lanka in 1974.



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Geological Structure

Introduction:

The **geological structure** of a country plays a crucial role in understanding the **types and characteristics of rocks and slopes, the physical and chemical properties of soils, the availability of minerals, and the surface and underground water resources**. In India, the geological structure has been classified into several rock systems, each reflecting the country's complex and varied geological history.

Geological Structure and Time Scale:

The term "**geological structure**" refers to the arrangement and deposition of rocks in the earth's crust, often resulting from or influenced by earth movements. It also encompasses the morphological features of rocks, such as the **Gondwana structure**. The Geological Time Scale, a chronological framework for dating various geological formations and life according to their origin, evolution, and extinction, was first developed by Giovanni Ardunia in 1760. The Standard Geological Time Scale, recognized internationally, was established during the International Geological Congress in 1881, Italy. **The Indian Geological Time Scale, advocated by T.S. Holland**, provides a specific timeline for geological formations in India.

India's geological history can be analyzed through its physiographic divisions, including **Peninsular region** and **Extra Peninsular region**. Significant events in India's geological history include the **formation of the Earth's crust**, the **upheaval of the Himalayas** during the Tertiary period, and the **aggradational formation of the Indo-Gangetic plain** during the Pleistocene period, which continues through sedimentation in the floodplains of the rivers and the lower Gangetic plain.

Major Rock Systems in India:

→ Not mentioned in NCERT

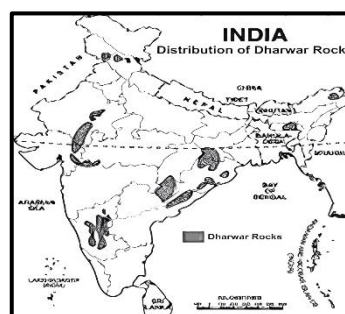
Based on this geological history, the **Geological Survey of India** classifies the country's rock systems into **four major divisions** - Archaean Rock System, Purana Rock System, Dravidian Rock System and Aryan Rock System.

Archaean Rock System (Pre-Cambrian Rocks):

The earliest phase of tectonic evolution marked the **cooling and solidification of the Earth's upper crust** during the Archaean era, which occurred over **2.5 billion years ago**. This system is represented by gneisses and granites, forming the core of the Indian Craton, a block of the Indian subcontinent that was once part of **Gondwanaland**.

The Archaean Rock System includes:

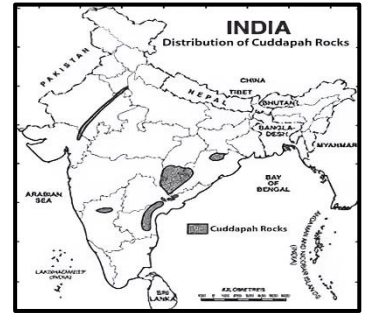
- **Archaean System** - Comprising granites and gneisses, these rocks are the oldest on Earth, formed through the solidification of molten magma. They are also known as the "**Basement Complex**," forming the base for new layers.
- **Dharwar System** - This system represents the **first sedimentary rocks** formed from the metamorphosis of sediments from Archaean gneisses and schists. These rocks are **highly metamorphosed, economically important** due to their mineral content, and found abundantly in **Karnataka's Dharwar district**.



Purana Rock System:

The Purana Rock System, consisting of the **Cuddapah and Vindhyan rock systems**, was formed between **1400-600 million years ago** through the **erosion and deposition of Archean and Dharwar rocks**. These rocks are primarily sedimentary.

- **Cuddapah System** - Named after the **Cuddapah district in Andhra Pradesh**, this system features sedimentary rocks like **sandstone and limestone**, deposited in synclinal folds. These rocks contain ores of iron, manganese, and other minerals.
- **Vindhyan System** - Deriving its name from the **Vindhyan mountains**, this system comprises ancient sedimentary rocks, including **diamond-bearing regions like Panna and Golconda**. Although devoid of metalliferous minerals, the system provides large quantities of durable stones and glass-making sand.



Dravidian Rock System (Palaeozoic):

Formed during the Paleozoic era (**600-300 million years ago**), the Dravidian Rock System primarily consists of sedimentary rocks found in **extra-Peninsular regions** like the Himalayas and Gangetic plain. The system marks the beginning of life on Earth, with rocks containing plenty of fossil evidence. The Dravidian period also saw the formation of **high-quality Carboniferous coal**, although these formations are **not abundant in India**. Key components of the Dravidian system include:

- **Carboniferous Rocks** - Formed around **350 million years ago**, these rocks primarily consist of **limestone, shale, and quartzite**. Notably, Mount Everest is composed of Upper Carboniferous limestone.

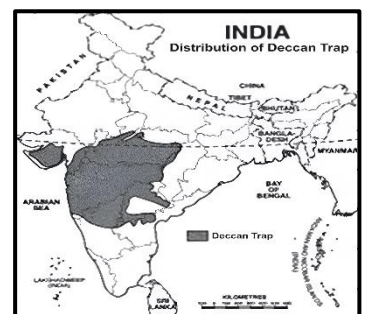
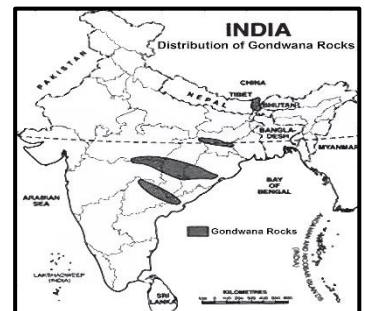
Aryan Rock System:

Beginning in the **Upper Carboniferous period**, the Aryan Rock System extends from the Upper Carboniferous to the Holocene period.

This Aryan system is classified into several groups:

Deccan Plateau
Andhra Pradesh
Kutch
Himalayan Region

- **Gondwana System** - Named after the **Gonds, an indigenous people of Telangana and Andhra Pradesh**, this system consists of deposits laid down in synclinal troughs on the ancient plateau surface since the **Permian period (250 million years ago)**. The Gondwana rocks contain nearly **98% of India's coal reserves**, along with other valuable minerals like iron ore and copper.
- **Jurassic System** - This system features marine transgressions that led to thick shallow water deposits in regions like **Rajasthan and Kachchh**, with coral limestone, sandstone, and shales.
- **Deccan Trap** - Resulting from volcanic outbursts over **Peninsular India** between the **Cretaceous and Eocene periods**, the Deccan Trap covers a vast area with basaltic lava flows. Weathering of these rocks has given rise to **black cotton soil** known as 'regur.'
- **Tertiary System** - The Tertiary period, spanning from the Eocene to the **Pliocene (60-7 million years ago)**, is significant in India's geological history as it saw the upliftment of the Tethys geosyncline, leading to the formation of the Himalayas. The Tertiary Succession is found in regions like the **Bengal and Ganges delta**, and the **outer Himalayan regions**.



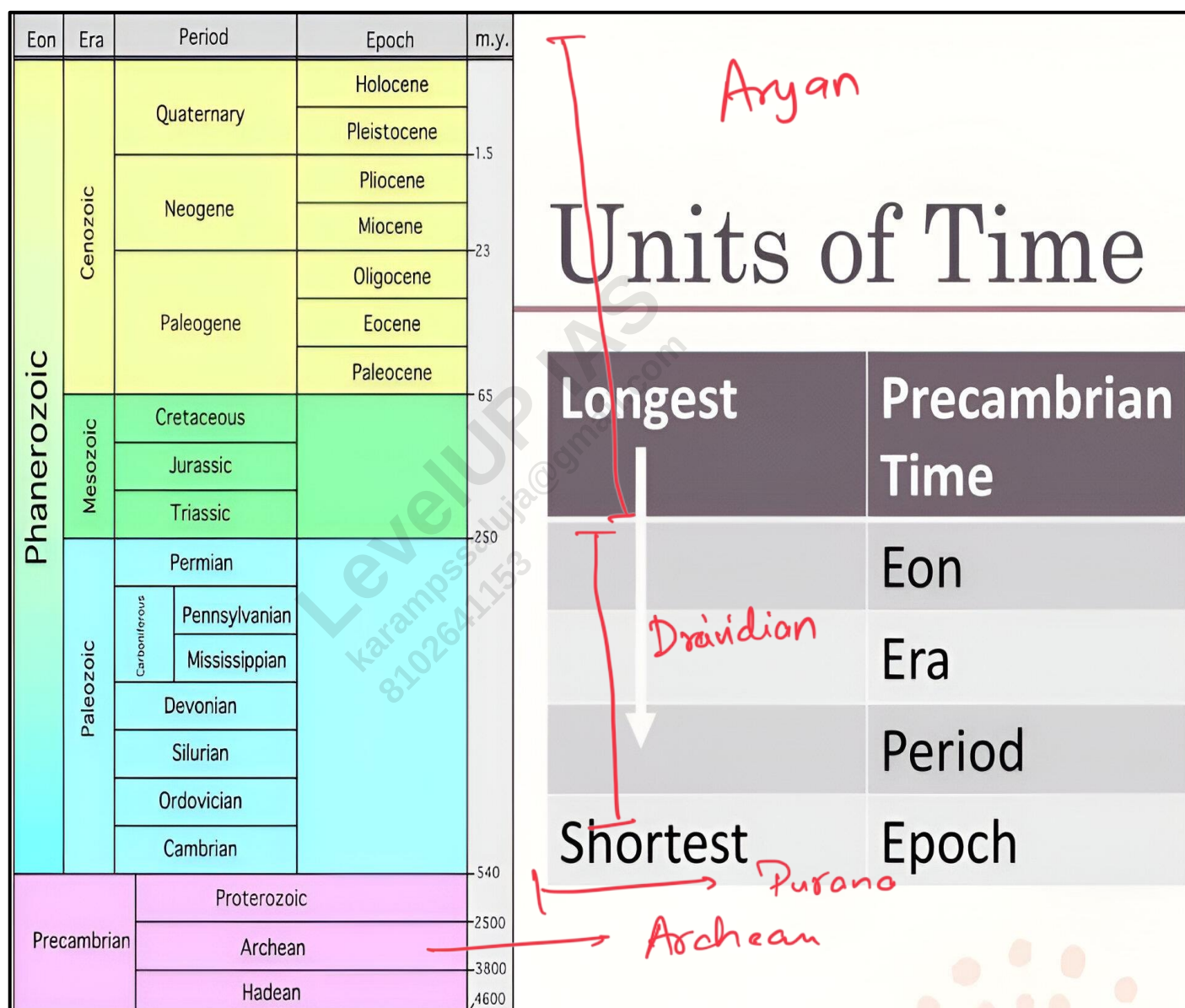
A Diverse and Unique History:

India's geological history is marked by its unique and diverse characteristics, with different regions containing rocks from various geologic periods. The Indian Craton was once a part of the supercontinent Pangaea. Following the **breakup** of the **Gondwanaland craton** around **225 million years ago** and its **movement toward the Eurasian craton**, the **Himalayan Mountain range** began to form **approximately 65 million years ago**.

In the aftermath of the **Outer Himalayas' formation**, **widespread alluvial plains** started developing during the **Upper Pliocene and Pleistocene epochs**, leading to the creation of the Indo-Ganga-Brahmaputra plain. As a result, different regions of India feature rocks representing nearly every type of geological structure from various geologic periods.

The stratigraphy of India can be categorized into several divisions, including the **Archean System**, **Dharwar System**, **Cuddapah System**, **Vindhyan System**, **Dravidian System**, and **Aryan System** (which includes the Gondwana, Jurassic, Deccan Trap, Tertiary, and Quaternary rocks).

India's geological strata span from the **oldest rocks to the newest alluvial** formations. The ancient rocks of the Archaean period are predominantly found in peninsular India. Sedimentary rocks are prevalent in areas formed by sediment deposition from the Indo-Gangetic plains. Additionally, various sedimentary and metamorphic rocks can be traced in the Great Folded Mountainous region.



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